

A Quantitative OSINT Model for Executive Protection Risk in Global Energy Operations

Country, city, support-access, and protective intelligence posture modeling for executive travel, site visits, corporate events, and energy-sector security planning

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Executive Summary

This report presents a quantitative OSINT model for estimating executive protection risk in global energy operations. The model combines ACLED event data, World Bank governance and macro indicators, a homicide-rate violent-crime proxy, energy-sector exposure indicators, recent risk momentum features, city/location-level protective intelligence, airport and medical support-access proxies, regional spillover analysis, Monte Carlo robustness testing, an executive-facing intelligence signal layer, and a trip-level Protective Intelligence posture model with a COA-oriented decision-support layer. The output is designed to support strategic thinking around executive travel, site visits, public events, operating-environment risk, and intelligence-led protective planning. The highest-ranked country in the current model output is Ukraine with an Executive Protection Risk Score of 69.76. The top Executive Protection Intelligence Signal is Ukraine, classified as High Attention with a signal score of 77.25. Baseline model maturity is classified as Fully Data-Driven, with 5 of 5 major baseline components populated. Forward layer initialized; target-year ACLED data unavailable, so forward scores are baseline-retained. Sensitivity analysis indicates: The top-risk group is meaningfully sensitive to weighting assumptions; interpret exact ranks with caution. The 2026 forward layer is included, but current-run 2026 ACLED YTD data was unavailable for the forward-country set; therefore, forward scores retain the calibrated 2024 baseline until valid 2026 ACLED data is available.

Key Findings

Finding	Current Output
Highest baseline EP risk country	Ukraine (69.76, Elevated)
Top intelligence signal	Ukraine (High Attention, 77.25)
Monte Carlo robustness	13 countries show at least 90% simulated top-20 probability
Highest regional spillover exposure	Ukraine (44.51)
Sensitivity interpretation	The top-risk group is meaningfully sensitive to weighting assumptions; interpret exact ranks with caution.
2026 forward layer	Target-year ACLED data unavailable; forward scores are baseline-retained
Top Protective Intelligence posture priority	TRIP001 / Salvador, Brazil (87.09, Severe PI Priority)
Top COA decision-support priority	TRIP001 / Salvador, Brazil (87.09, Senior Review - Senior security review)

These findings should be interpreted as strategic screening outputs rather than tactical protective decisions.

Model Framework

The Executive Protection Risk Score is constructed from five weighted components: civil unrest and political violence, governance and rule-of-law risk, violent-crime proxy risk, energy-sector exposure, and recent risk momentum. The civil unrest component incorporates event volume, fatality intensity, event composition, geographic spread, coordinate spread, and high-relevance ACLED sub-event types.

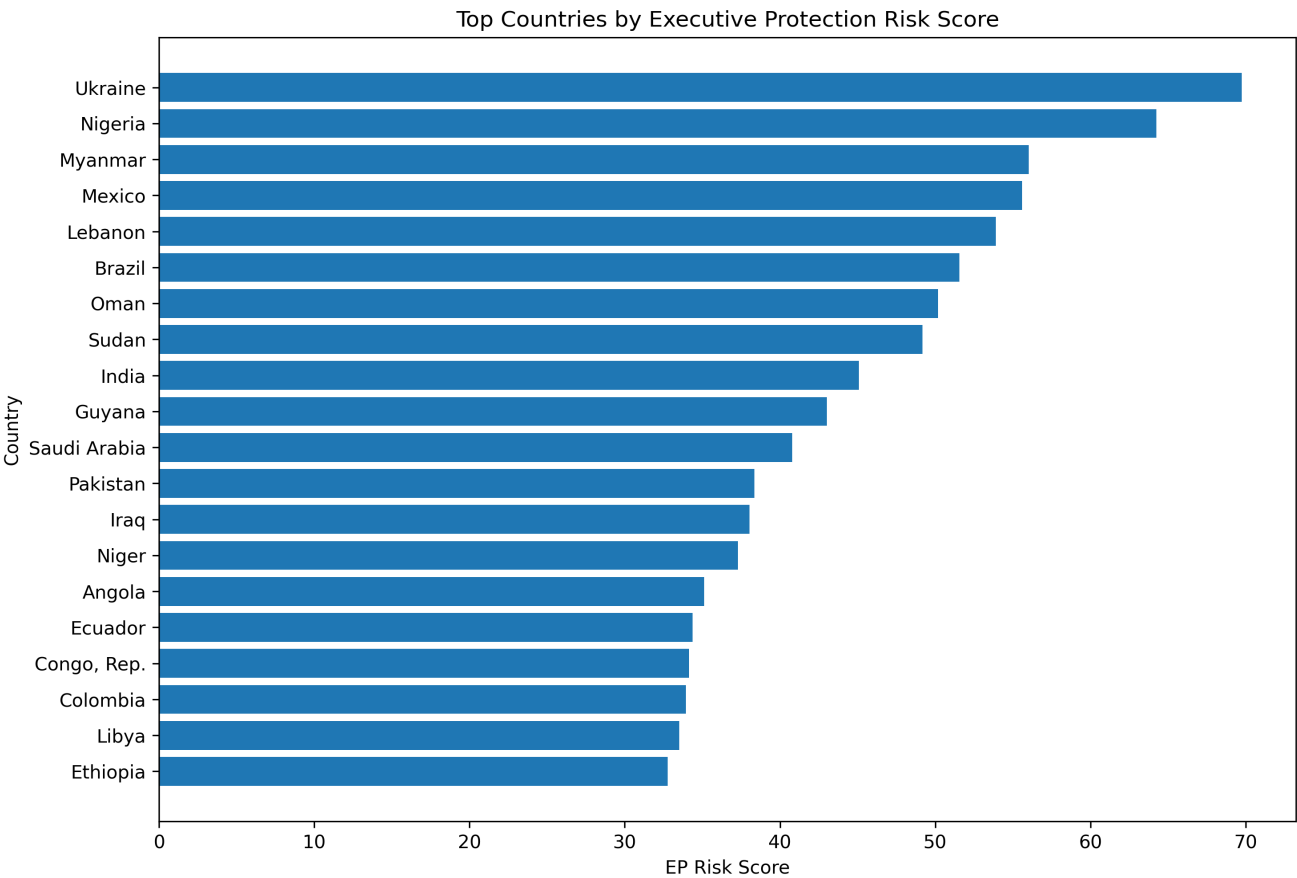
After the weighted component score is calculated, the model applies a bounded severity calibration layer. This uplift is designed to reduce score compression in countries with extreme conflict exposure, high fatality intensity, broad geographic spread of violence, fragile governance combined with violence, strategic energy exposure combined with instability, or elevated recent momentum. The

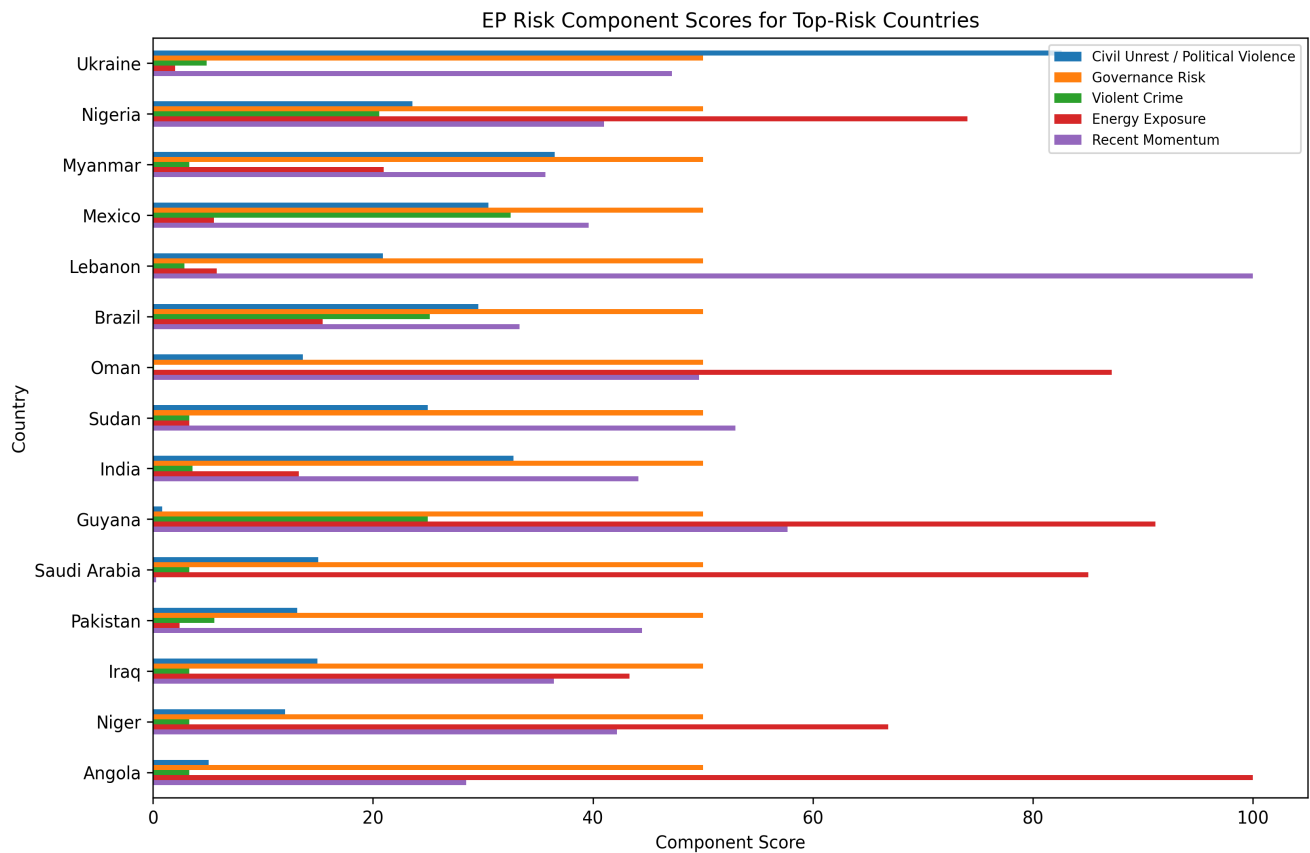
calibration is capped so that it improves operational interpretability without overwhelming the underlying data-driven component scores.

The model does not represent any internal corporate security methodology. It is an open-source analytical framework intended for portfolio, research, and strategic risk-intelligence purposes.

Top Countries by Executive Protection Risk

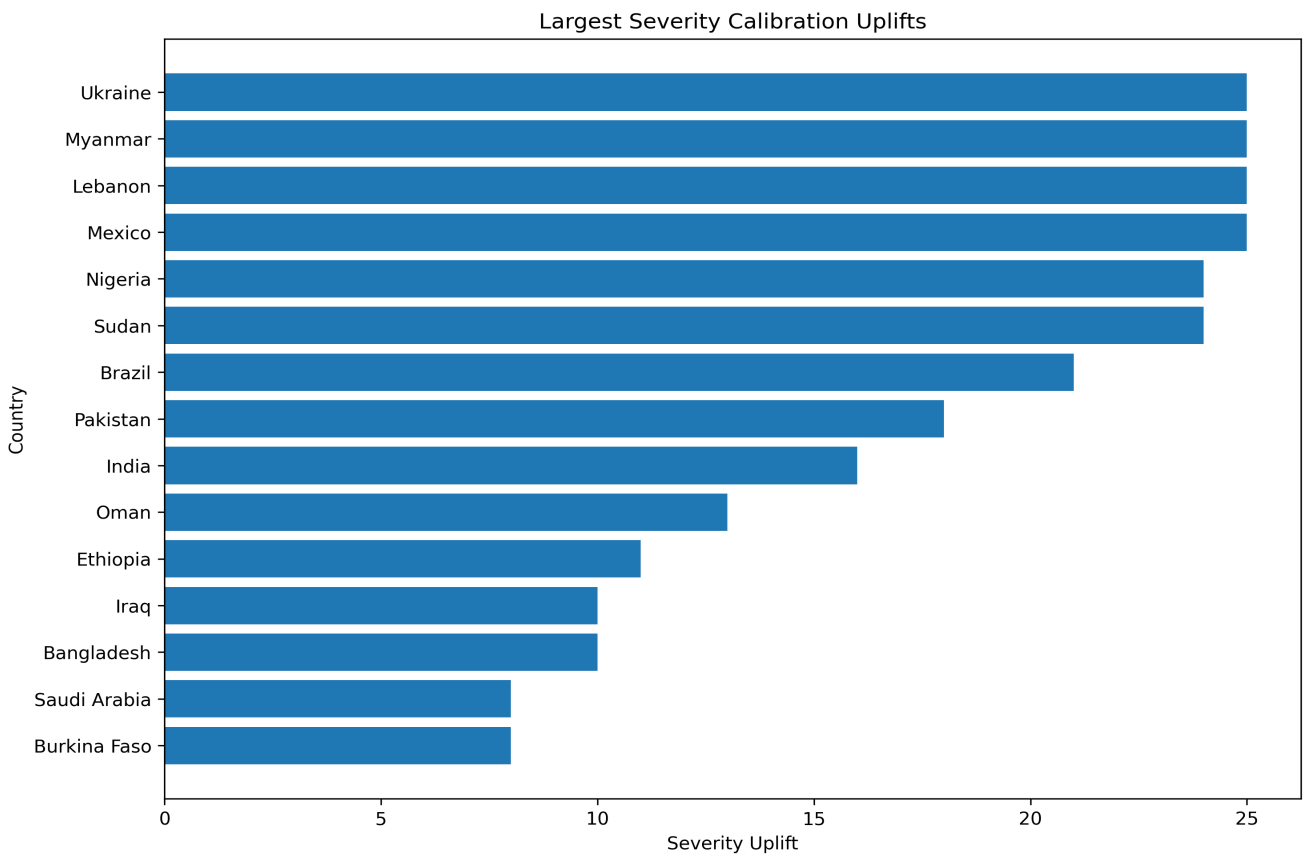
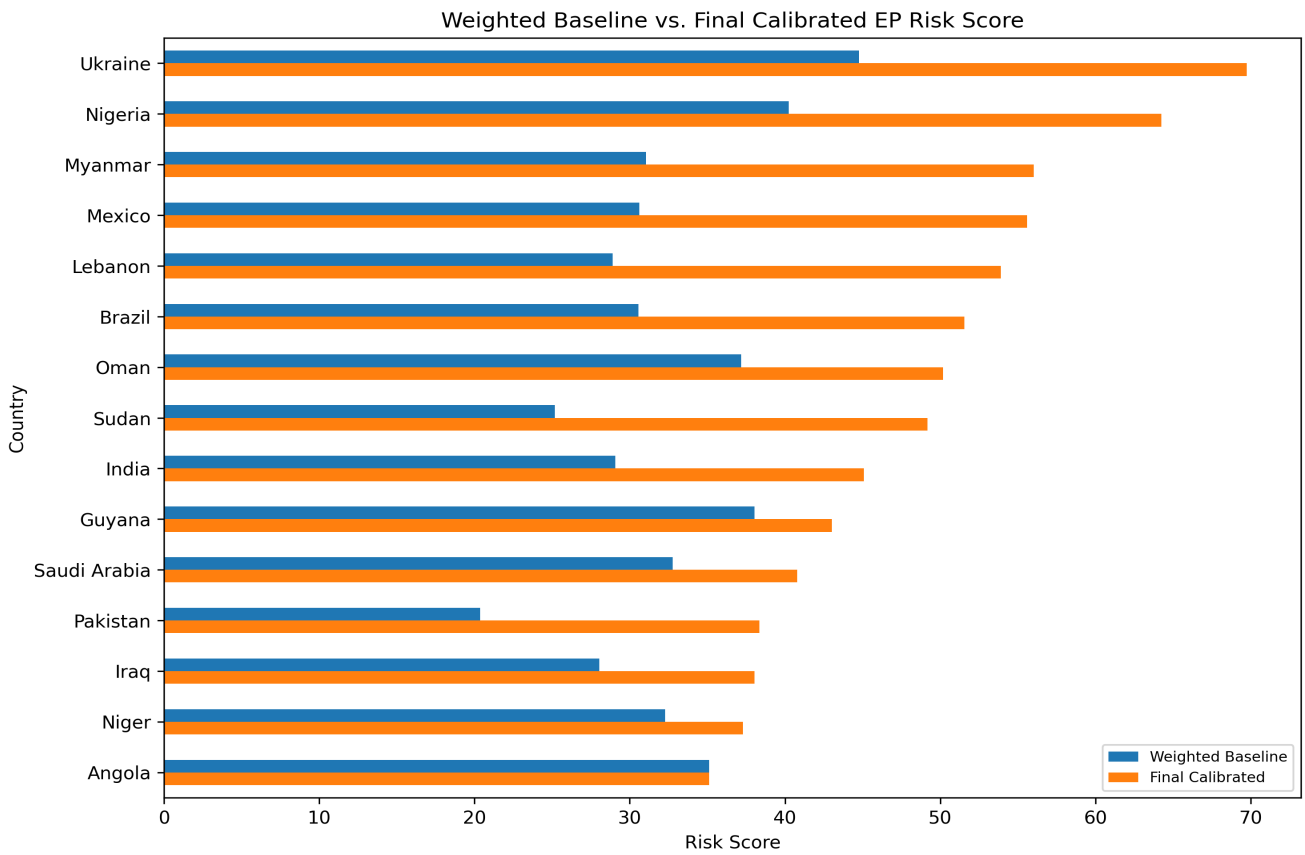
Country	EP Score	Bucket	Unrest	Gov.	Crime	Energy	Momentum
Ukraine	69.76	Elevated	82.61	50	4.86	2.0	47.2
Nigeria	64.25	Elevated	23.6	50	20.56	74.04	41.01
Myanmar	56.04	Elevated	36.53	50	3.29	20.96	35.66
Mexico	55.61	Elevated	30.48	50	32.5	5.53	39.61
Lebanon	53.9	Elevated	20.89	50	2.85	5.8	100.0
Brazil	51.55	Elevated	29.59	50	25.18	15.42	33.31
Oman	50.18	Elevated	13.62	50	0.09	87.19	49.66
Sudan	49.18	Moderate	24.97	50	3.29	3.29	52.94
India	45.08	Moderate	32.78	50	3.6	13.26	44.14
Guyana	43.03	Moderate	0.83	50	24.98	91.13	57.68
Saudi Arabia	40.78	Moderate	15.01	50	3.29	85.04	0.28
Pakistan	38.35	Moderate	13.11	50	5.59	2.41	44.45





Severity Calibration Layer

The severity calibration layer separates the initial weighted model score from the final calibrated Executive Protection Risk Score. This helps the model recognize countries where extreme conflict intensity, fatality concentration, geographic spread, compound governance weakness, or strategic energy exposure creates an operationally more severe environment than the weighted component score alone would suggest.

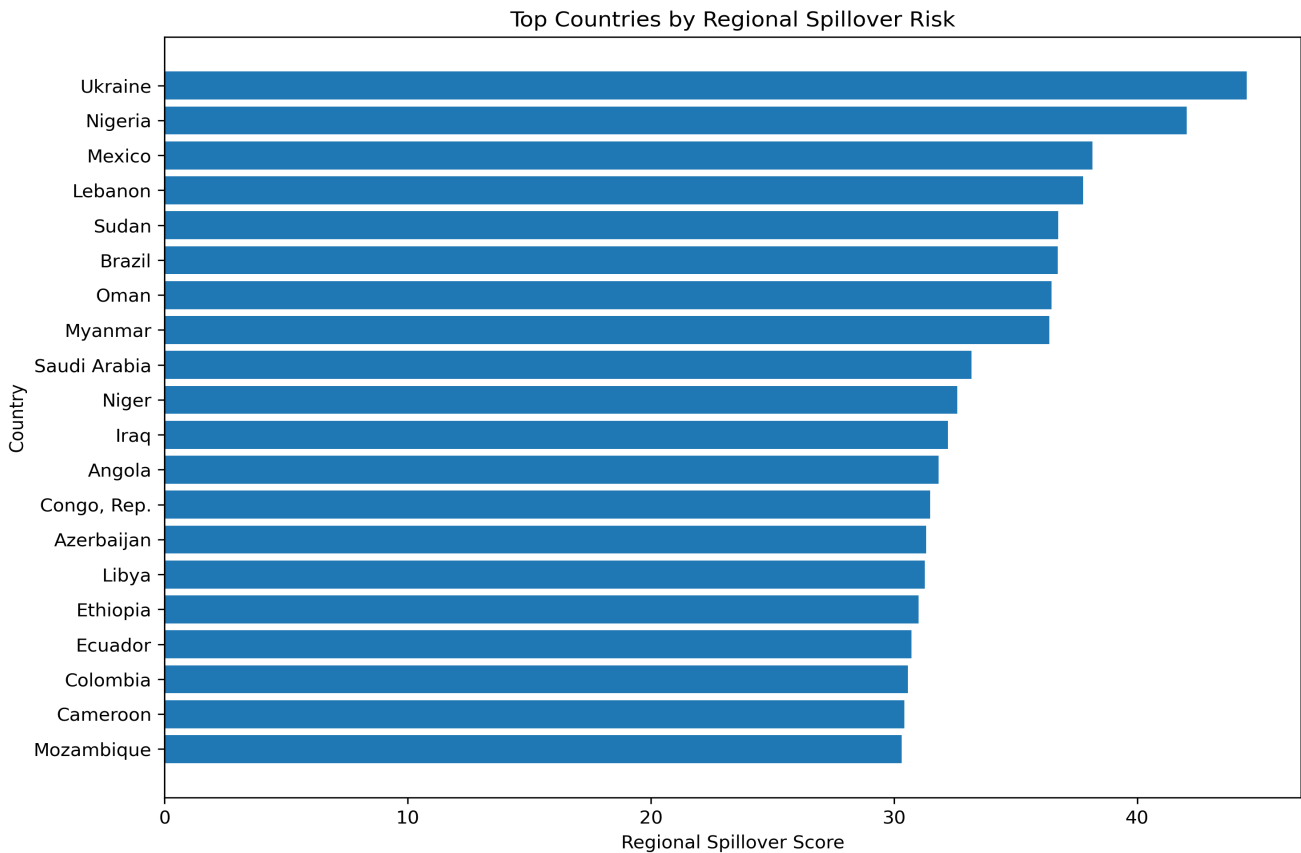


Country	Weighted	Uplift	Final	Bucket	Conflict	Fatality	Spread	Gov/Viol.	Energy	Momentum
Ukraine	44.76	25.0	69.76	Elevated	18.0	8.0	6.0	0.0	0.0	0.0
Nigeria	40.25	24.0	64.25	Elevated	10.0	8.0	6.0	0.0	0.0	0.0
Myanmar	31.04	25.0	56.04	Elevated	18.0	8.0	6.0	0.0	0.0	0.0
Mexico	30.61	25.0	55.61	Elevated	18.0	5.0	6.0	0.0	0.0	0.0
Lebanon	28.9	25.0	53.9	Elevated	18.0	5.0	0.0	0.0	0.0	5.0
Brazil	30.55	21.0	51.55	Elevated	18.0	0.0	3.0	0.0	0.0	0.0
Oman	37.18	13.0	50.18	Elevated	0.0	0.0	0.0	0.0	8.0	5.0
Sudan	25.18	24.0	49.18	Moderate	10.0	8.0	3.0	0.0	0.0	3.0
India	29.08	16.0	45.08	Moderate	10.0	0.0	6.0	0.0	0.0	0.0
Guyana	38.03	5.0	43.03	Moderate	0.0	0.0	0.0	0.0	0.0	5.0

The uplift is bounded and additive. It should be interpreted as a calibration adjustment for strategic screening, not as a standalone threat assessment.

Regional Spillover Risk

Regional spillover analysis evaluates whether a country's executive protection risk is amplified by elevated risk in nearby or analytically connected regional environments. This is useful for energy operations because executive travel, site visits, logistics, security posture, and regional operating assumptions are often influenced by instability beyond a single country border.



Top Regional Spillover Countries

Country	Region	EP	Bucket	Spillover	Flag	Reg Avg	Elev+	High/Sev
Ukraine	Europe / Eurasia	69.76	Elevated	44.51	Moderate Regional Spillover Exposure	26.55	1	0
Nigeria	Africa	64.25	Elevated	42.03	Moderate Regional Spillover Exposure	31.66	1	0
Mexico	Latin America	55.61	Elevated	38.15	Moderate Regional Spillover Exposure	28.04	2	0
Lebanon	Middle East	53.9	Elevated	37.78	Moderate Regional Spillover Exposure	30.72	2	0
Sudan	Africa	49.18	Moderate	36.75	Moderate Regional Spillover Exposure	31.66	1	0
Brazil	Latin America	51.55	Elevated	36.73	Moderate Regional Spillover Exposure	28.04	2	0
Oman	Middle East	50.18	Elevated	36.48	Moderate Regional Spillover Exposure	30.72	2	0
Myanmar	Southeast Asia	56.04	Elevated	36.39	Moderate Regional Spillover Exposure	22.34	1	0
Saudi Arabia	Middle East	40.78	Moderate	33.19	Moderate Regional Spillover Exposure	30.72	2	0
Niger	Africa	37.28	Moderate	32.59	Moderate Regional Spillover Exposure	31.66	1	0
Iraq	Middle East	38.03	Moderate	32.22	Moderate Regional Spillover Exposure	30.72	2	0
Angola	Africa	35.12	Moderate	31.83	Moderate Regional Spillover Exposure	31.66	1	0

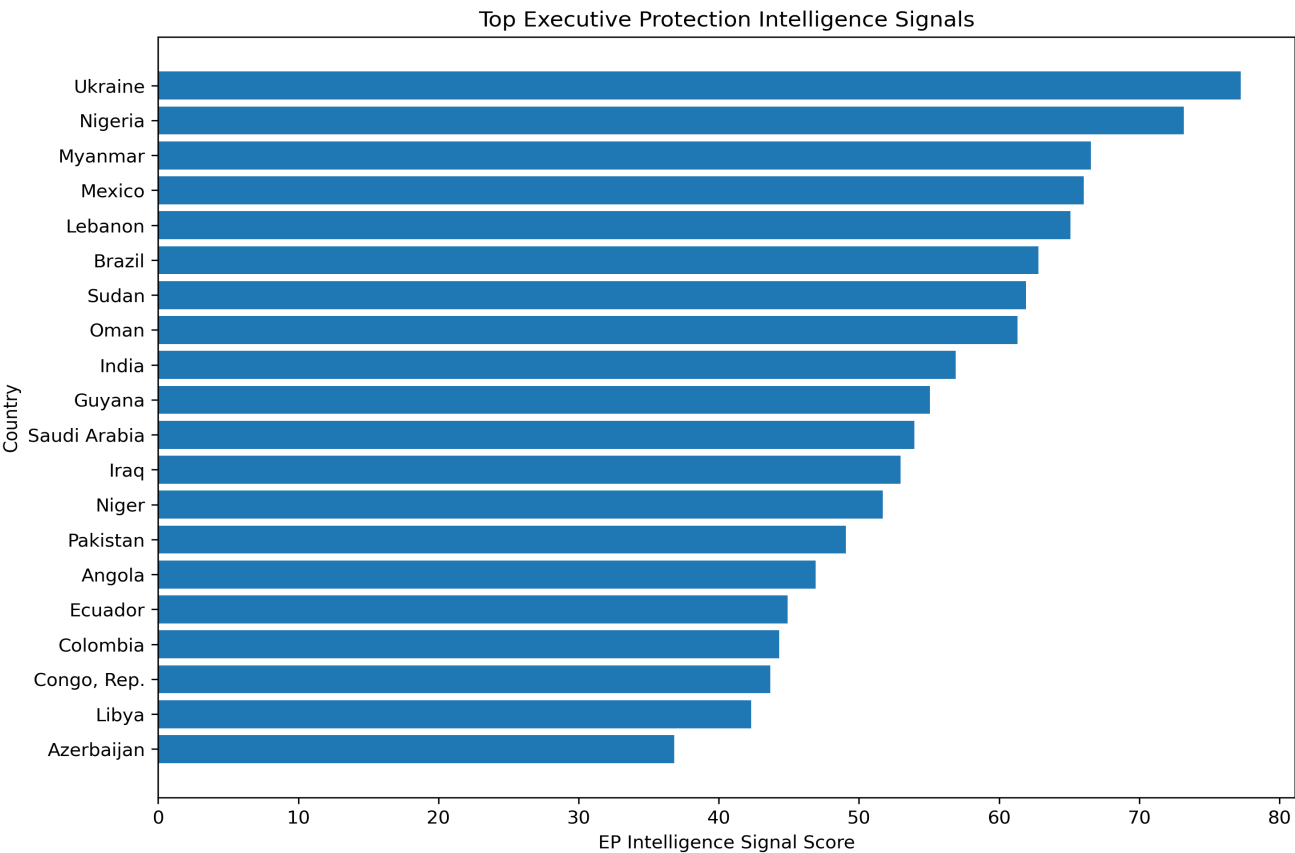
Regional Context Notes

Country	Regional Context
Ukraine	Europe / Eurasia has 1 elevated-or-higher countries and 0 high/severe countries in the current model universe.
Nigeria	Africa has 1 elevated-or-higher countries and 0 high/severe countries in the current model universe.
Mexico	Latin America has 2 elevated-or-higher countries and 0 high/severe countries in the current model universe.
Lebanon	Middle East has 2 elevated-or-higher countries and 0 high/severe countries in the current model universe.
Sudan	Africa has 1 elevated-or-higher countries and 0 high/severe countries in the current model universe.
Brazil	Latin America has 2 elevated-or-higher countries and 0 high/severe countries in the current model universe.

Regional spillover scores should be interpreted as a strategic screening overlay. They do not imply direct contagion or causal transmission of risk from one country to another.

Executive Protection Intelligence Signal

The Executive Protection Intelligence Signal combines the baseline EP risk score, scenario pressure, Monte Carlo top-20 stability, regional spillover exposure, and forward-risk pressure into a single executive-facing monitoring signal. The goal is to translate model outputs into a concise analyst interpretation suitable for strategic screening and portfolio presentation.



Top Executive Protection Intelligence Signals

Country	Region	EP	Bucket	Signal Score	Signal	Scenario	MC Stable	Spillover
Ukraine	Europe / Eurasia	69.76	Elevated	77.25	High Attention	94.22	100.0	44.51
Nigeria	Africa	64.25	Elevated	73.18	High Attention	86.78	100.0	42.03
Myanmar	Southeast Asia	56.04	Elevated	66.56	Elevated Monitoring	75.7	100.0	36.39
Mexico	Latin America	55.61	Elevated	66.05	Elevated Monitoring	75.11	100.0	38.15
Lebanon	Middle East	53.9	Elevated	65.09	Elevated Monitoring	72.8	100.0	37.78
Brazil	Latin America	51.55	Elevated	62.82	Elevated Monitoring	69.63	100.0	36.73
Sudan	Africa	49.18	Moderate	61.91	Elevated Monitoring	66.43	100.0	36.75
Oman	Middle East	50.18	Elevated	61.33	Elevated Monitoring	67.78	100.0	36.48
India	South Asia	45.08	Moderate	56.92	Elevated Monitoring	60.89	99.9	30.0
Guyana	Other / Unmapped	43.03	Moderate	55.07	Elevated Monitoring	58.12	100.0	26.76
Saudi Arabia	Middle East	40.78	Moderate	53.95	Routine Monitoring	55.08	99.3	33.19
Iraq	Middle East	38.03	Moderate	52.96	Routine Monitoring	51.37	99.8	32.22

Analyst Priority Notes

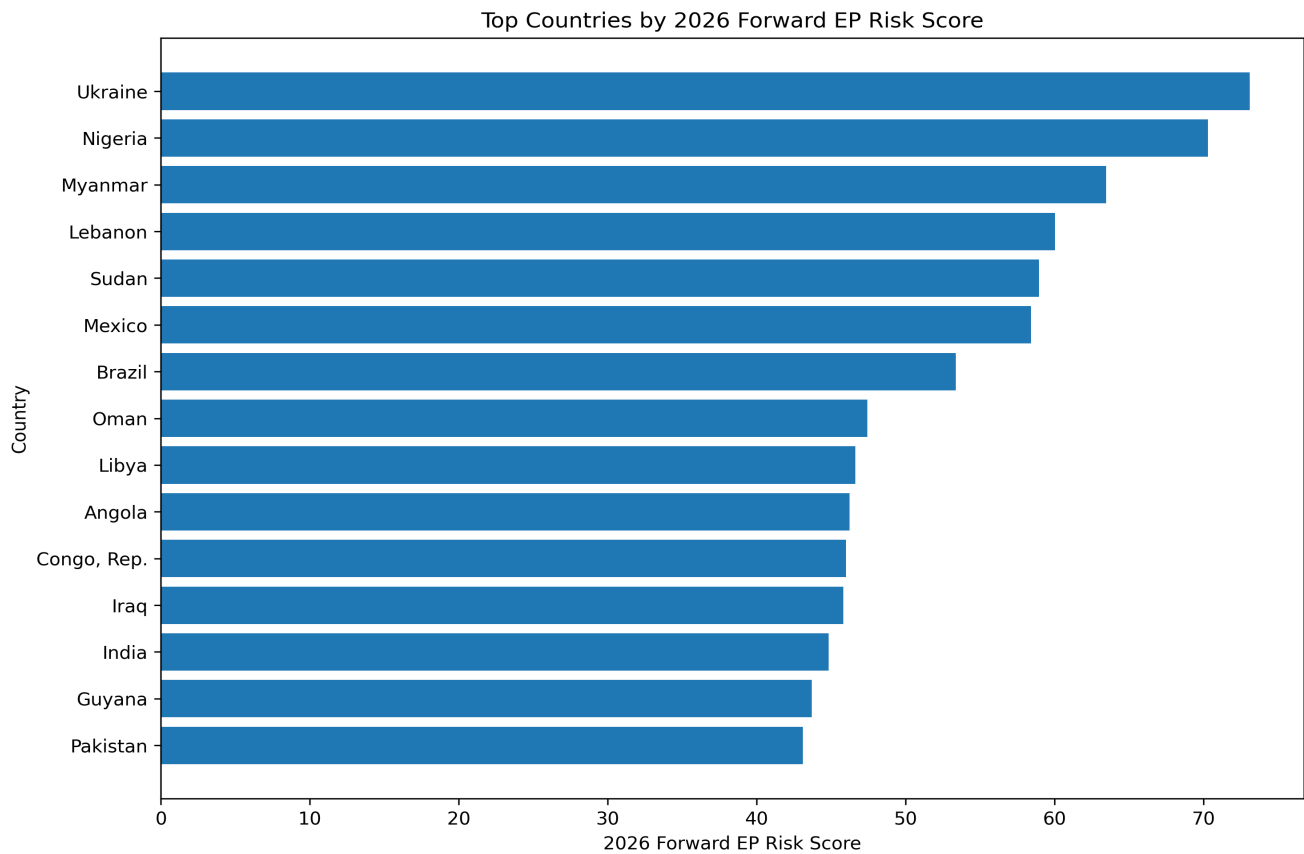
Country	Priority Note
Ukraine	High Attention: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.
Nigeria	High Attention: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.
Myanmar	Elevated Monitoring: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.
Mexico	Elevated Monitoring: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.
Lebanon	Elevated Monitoring: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.
Brazil	Elevated Monitoring: baseline bucket is Elevated; regional context is Moderate Regional Spillover Exposure; Monte Carlo classification is Highly Stable Top-20 Risk; Forward risk stable / mixed.

The intelligence signal is a composite screening indicator. It should be interpreted as an analyst prioritization layer, not as a tactical travel approval decision or real-world event probability forecast.

2026 Forward Risk Update

The 2026 Forward Risk Update is designed as a nowcast-style layer built on top of the calibrated 2024 baseline model. In the current run, 2026 ACLED YTD data was unavailable for the forward-country set. As a result, the model retained the calibrated 2024 baseline rather than interpreting missing event data as risk easing. The forward layer is configured to compare 2026-01-01 to 2026-05-20 against the same-period comparison window of 2025-01-01 to 2025-05-20.

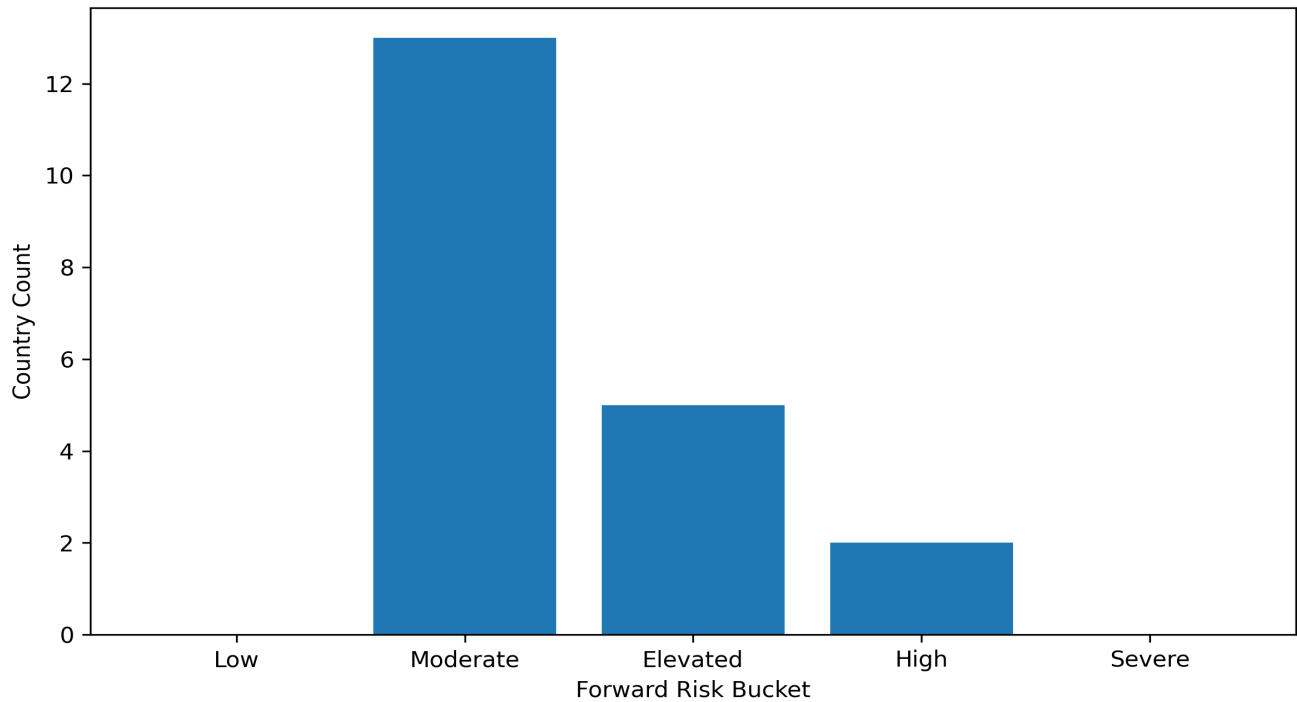
The forward scores below should therefore be read as baseline-retained screening scores, not as a true 2026 risk forecast.



Top 2026 Forward EP Risk Scores

Country	2024 Base	Base	2026	Forward	Change	Flag	Target Data
Ukraine	73.1	High	73.1	High	0.0	Stable / mixed	Target-year ACLED data unavailable
Nigeria	70.31	High	70.31	High	0.0	Stable / mixed	Target-year ACLED data unavailable
Myanmar	63.47	Elevated	63.47	Elevated	0.0	Stable / mixed	Target-year ACLED data unavailable
Lebanon	60.02	Elevated	60.02	Elevated	0.0	Stable / mixed	Target-year ACLED data unavailable
Sudan	58.96	Elevated	58.96	Elevated	0.0	Stable / mixed	Target-year ACLED data unavailable
Mexico	58.43	Elevated	58.43	Elevated	0.0	Stable / mixed	Target-year ACLED data unavailable
Brazil	53.38	Elevated	53.38	Elevated	0.0	Stable / mixed	Target-year ACLED data unavailable
Oman	47.42	Moderate	47.42	Moderate	0.0	Stable / mixed	Target-year ACLED data unavailable
Libya	46.63	Moderate	46.63	Moderate	0.0	Stable / mixed	Target-year ACLED data unavailable
Angola	46.25	Moderate	46.25	Moderate	0.0	Stable / mixed	Target-year ACLED data unavailable

2026 Forward EP Risk Bucket Distribution



Forward Data Availability Notes

Country	Forward	Change	Flag	Note	Fetch Status
Ukraine	73.1	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Nigeria	70.31	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Myanmar	63.47	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Lebanon	60.02	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Sudan	58.96	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Mexico	58.43	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Brazil	53.38	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable
Oman	47.42	0.0	Stable / mixed	2026 ACLED data unavailable; baseline retained	Target-year ACLED data unavailable

Run-to-Run Change Detection

Run-to-run change detection compares the current Executive Protection Risk rankings against the previous saved model snapshot. This helps identify countries where modeled risk increased, decreased, changed risk bucket, or moved materially in the country rankings.

No material run-to-run changes were detected in this model run. Current rankings, scores, and risk buckets matched the previous saved snapshot within the model's change-detection thresholds.

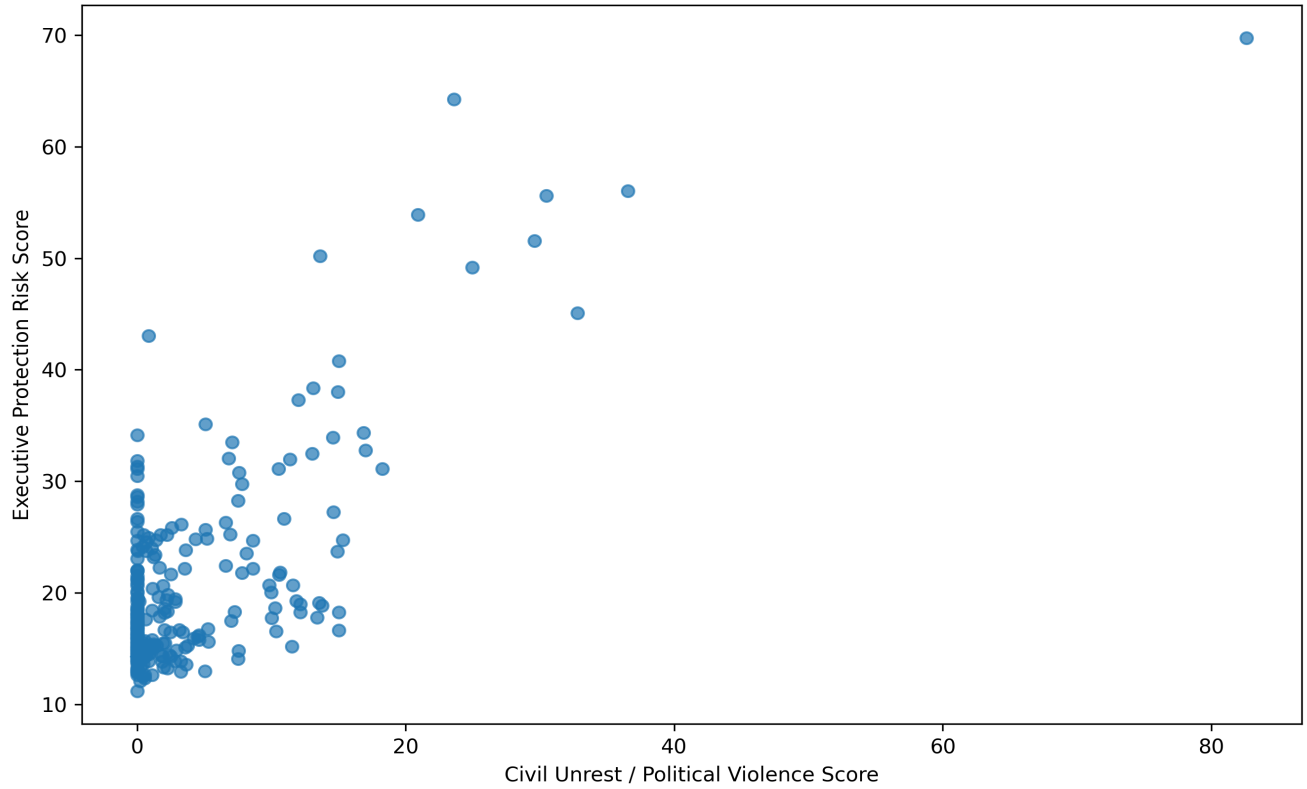
The snapshot was still refreshed, so future runs can identify new movement when input data, model parameters, or forward-risk signals change.

Civil Unrest and Political Violence

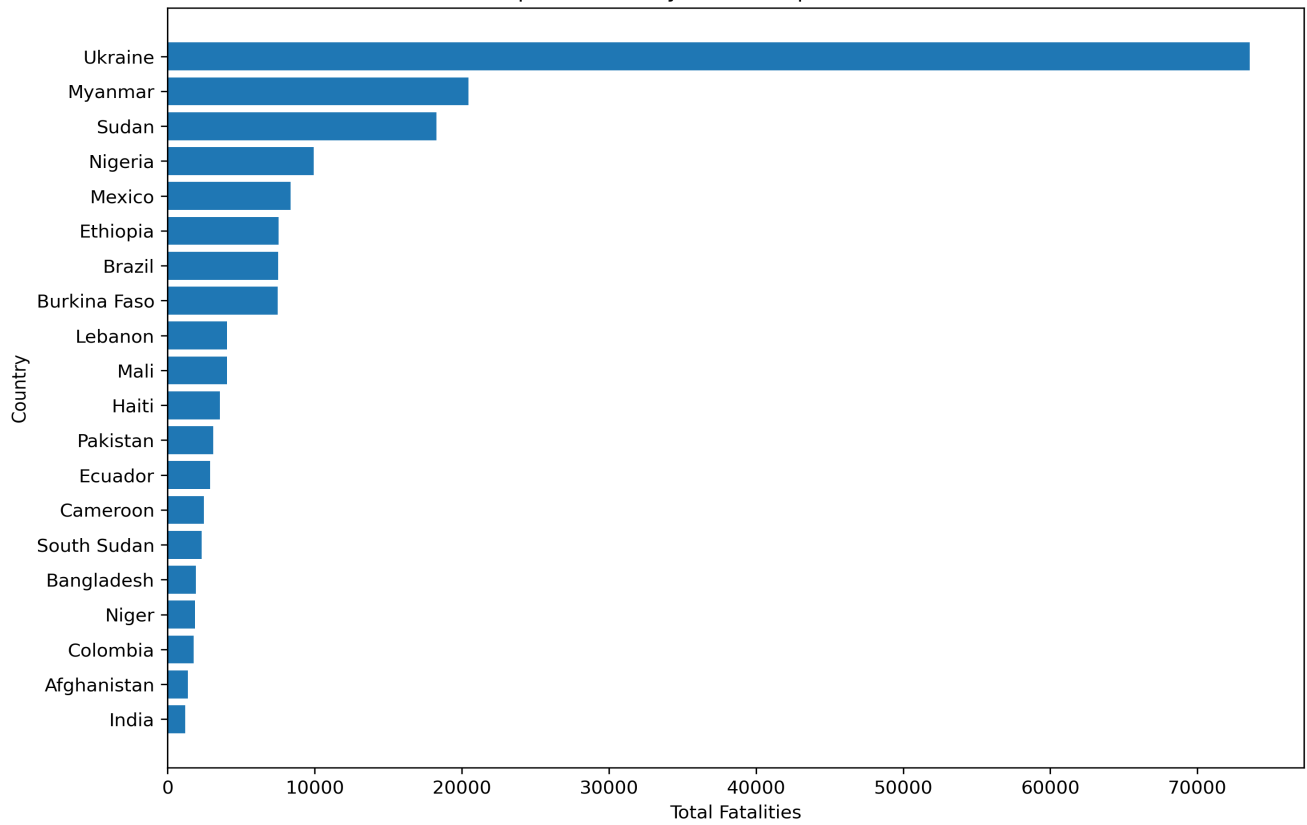
ACLED event data is used to estimate the security environment relevant to executive movement, site visits, public events, and route planning. The model uses protests, riots, battles, remote violence, violence against civilians, fatalities, high-fatality events, geographic spread, and coordinate-pair spread to separate isolated event activity from broader operating risk.

Country	Events	Unrest	Violent	Fatalities	Fatal Events	High-Fatality
Ukraine	53557.0	345.0	53212.0	73572.0	11018.0	2855.0
Nigeria	5349.0	1255.0	4094.0	9917.0	2627.0	495.0
Myanmar	14292.0	1207.0	13085.0	20457.0	4506.0	1043.0
Mexico	13564.0	6299.0	7265.0	8353.0	5417.0	172.0
Lebanon	13693.0	307.0	13386.0	4050.0	1164.0	198.0
Brazil	9918.0	1482.0	8436.0	7500.0	6164.0	33.0
Oman	5.0	0.0	5.0	10.0	3.0	1.0
Sudan	6046.0	81.0	5965.0	18287.0	2539.0	924.0
India	25731.0	24124.0	1607.0	1204.0	783.0	33.0
Guyana	83.0	80.0	3.0	2.0	2.0	0.0

Civil Unrest / Political Violence vs. EP Risk



Top Countries by ACLED-Reported Fatalities



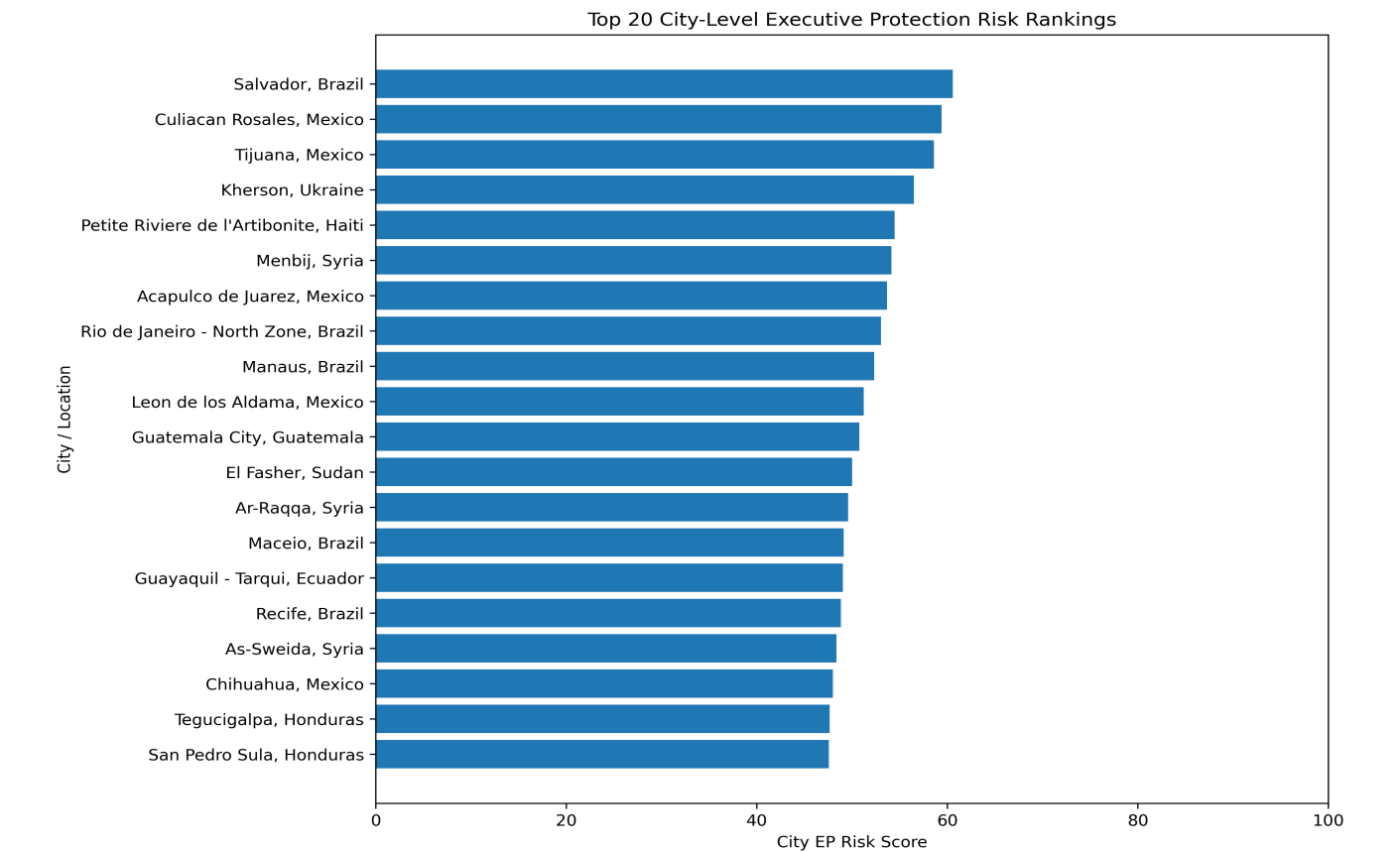
City-Level Protective Intelligence Layer

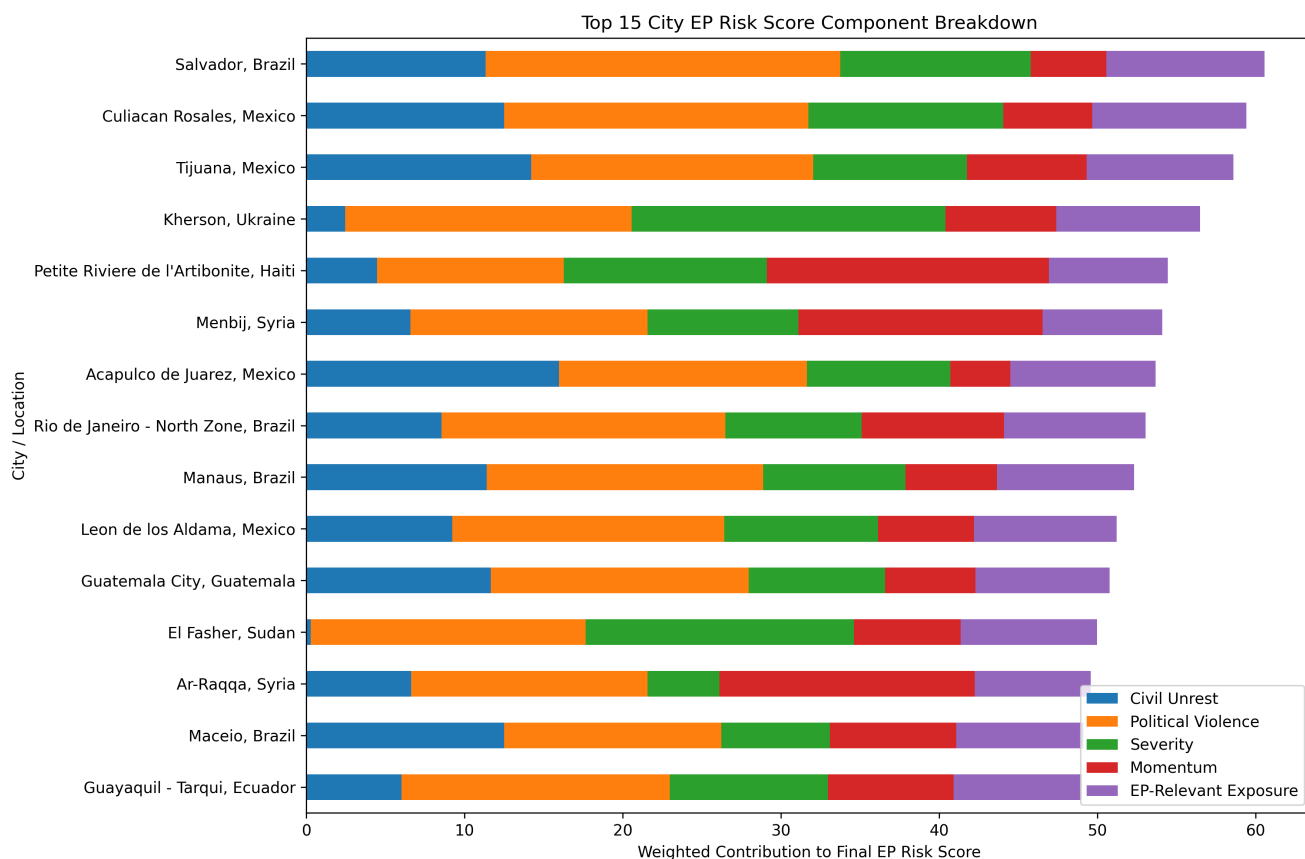
The city-level protective intelligence layer extends the baseline country model by ranking ACLED city/location records according to recent civil unrest, political violence, severity, event momentum, and EP-relevant exposure. This layer is designed for strategic screening and travel-intelligence prioritization, not tactical route planning.

The highest-ranked city/location in the current ACLED city layer is Salvador, Brazil, with a City EP Risk Score of 60.56 and a signal of Elevated.

Top City/Location EP Risk Rankings

Rank	City/Location	Country	Admin1	90d Events	90d Fatal.	Unrest	Violence	Severity	Momentum	EP Rel.	City EP	Signal	Driver
1	Salvador	Brazil	Bahia	242.0	199.0	45.3	89.64	60.22	23.92	99.97	60.56	Elevated	EP-Relevant Exposure
2	Culiacan Rosales	Mexico	Sinaloa	199.0	169.0	49.96	76.94	61.64	28.05	97.5	59.41	Elevated	EP-Relevant Exposure
3	Tijuana	Mexico	Baja California	131.0	97.0	56.81	71.36	48.51	37.88	92.7	58.59	Elevated	EP-Relevant Exposure
4	Kherson	Ukraine	Kherson	204.0	7870.0	9.82	72.39	99.22	35.06	90.82	56.49	Elevated	Severity
5	Petite Riviere de l'Artibonite	Haiti	Artibonite	30.0	450.0	17.89	47.19	64.2	89.11	75.24	54.45	Elevated	Recent Momentum
6	Menbij	Syria	Aleppo	41.0	94.0	26.33	59.91	47.67	77.2	75.76	54.11	Elevated	Recent Momentum
7	Acapulco de Juarez	Mexico	Guerrero	122.0	54.0	63.82	62.68	45.34	18.96	91.83	53.67	Elevated	EP-Relevant Exposure
8	Rio de Janeiro - North Zone	Brazil	Rio de Janeiro	100.0	47.0	34.13	71.78	43.17	44.93	89.55	53.05	Elevated	EP-Relevant Exposure
9	Manaus	Brazil	Amazonas	78.0	66.0	45.58	69.94	44.96	28.93	86.65	52.32	Elevated	EP-Relevant Exposure
10	Leon de los Aldama	Mexico	Guanajuato	105.0	99.0	36.88	68.75	48.68	30.34	90.16	51.23	Elevated	EP-Relevant Exposure
11	Guatemala City	Guatemala	Guatemala	67.0	59.0	46.58	65.28	43.08	28.52	84.86	50.77	Elevated	EP-Relevant Exposure
12	El Fasher	Sudan	North Darfur	115.0	1390.0	1.07	69.54	84.85	33.71	86.22	49.99	Moderate	EP-Relevant Exposure





An interactive geospatial city-risk map is generated separately as an HTML dashboard artifact: [city_ep_risk_map.html](#). The PDF summarizes the ranked output and chart views, while the Streamlit dashboard provides the interactive map experience.

Methodology note: ACLED location names are treated as city/location-level records. Some entries may represent neighborhoods, towns, villages, districts, or event locations rather than formal city boundaries.

Airport and Medical Access Proxy Layer

The access-support proxy layer adds a planning-support view to the city-level model. It combines airport access, country-level medical capacity proxies, support access scoring, and support gap scoring to estimate where protective planning may face greater operating constraints. This layer is not a medical assessment, evacuation plan, or tactical protective operations product.

The highest access-adjusted operational EP risk location is Salvador, Brazil, with an operational EP risk score of 58.64. The support gap score is 52.88.

Top Access-Adjusted Operational City Rankings

Rank	City	Country	Admin1	City EP	IATA	Airport km	Airport	Medical	Gap	Operational	Signal
1	Salvador	Brazil	Bahia	60.56	SSA	21.56	66.78	17.63	52.88	58.64	Elevated Op. Concern
2	Culiacan Rosales	Mexico	Sinaloa	59.41	CUL	9.54	66.74	15.38	53.8	58.01	Elevated Op. Concern
3	Tijuana	Mexico	Baja California	58.59	TIJ	4.69	78.74	15.38	46.6	55.59	Elevated Op. Concern
4	Petite Riviere de l'Artibonite	Haiti	Artibonite	54.45	PAP	63.68	64.42	12.21	56.46	54.95	Elevated Op. Concern
5	Menbij	Syria	Aleppo	54.11	GZT	62.98	63.95	10.73	57.34	54.92	Elevated Op. Concern
6	Kherson	Ukraine	Kherson	56.49	KHE	8.79	66.78	32.09	47.1	54.14	Elevated Op. Concern
7	Acapulco de Juarez	Mexico	Guerrero	53.67	ACA	18.37	66.33	15.38	54.05	53.76	Elevated Op. Concern
8	El Fasher	Sudan	North Darfur	49.99	ELF	3.62	67.01	2.28	58.88	52.21	Elevated Op. Concern
9	Ar-Raqqa	Syria	Ar Raqqa	49.59	DEZ	125.17	59.38	10.73	60.08	52.21	Elevated Op. Concern
10	Manaus	Brazil	Amazonas	52.32	PLL	6.52	69.06	17.63	51.51	52.12	Elevated Op. Concern
11	Guatemala City	Guatemala	Guatemala	50.77	GUA	4.51	68.33	7.23	56.11	52.1	Elevated Op. Concern
12	Leon de los Aldama	Mexico	Guanajuato	51.23	BJX	25.5	66.93	15.38	53.69	51.84	Elevated Op. Concern

Note: the summary table uses IATA code and shortened operational signal labels to preserve PDF readability. Full airport names and status labels are provided in the access proxy detail table below.

Access Proxy Detail

City	Country	IATA	Airport Status	Beds/1k	Phys./1k	Medical Status	Operational Signal
Salvador	Brazil	SSA	Strong airport access	2.52	2.36	Low medical capacity proxy	Elevated Op. Concern
Culiacan Rosales	Mexico	CUL	Strong airport access	1.01	2.59	Low medical capacity proxy	Elevated Op. Concern
Tijuana	Mexico	TIJ	Strong airport access	1.01	2.59	Low medical capacity proxy	Elevated Op. Concern
Petite Riviere de l'Artibonite	Haiti	PAP	Moderate airport access	4.9	0.29	Low medical capacity proxy	Elevated Op. Concern
Menbij	Syria	GZT	Moderate airport access	1.43	1.52	Low medical capacity proxy	Elevated Op. Concern
Kherson	Ukraine	KHE	Strong airport access	6.14	3.53	Limited medical capacity proxy	Elevated Op. Concern
Acapulco de Juarez	Mexico	ACA	Strong airport access	1.01	2.59	Low medical capacity proxy	Elevated Op. Concern
El Fasher	Sudan	ELF	Strong airport access	0.66	0.25	Low medical capacity proxy	Elevated Op. Concern
Ar-Raqqa	Syria	DEZ	Limited airport access	1.43	1.52	Low medical capacity proxy	Elevated Op. Concern
Manaus	Brazil	PLL	Strong airport access	2.52	2.36	Low medical capacity proxy	Elevated Op. Concern

Interpretation note: the access-adjusted operational EP score increases when city-level risk is high and support access is constrained. Airport access is based on distance to large and medium airports where reference data is available. Medical capacity uses country-level World Bank proxy indicators and should not be interpreted as local hospital capability.

Protective Intelligence Exposure and Decision-Support Layer

The Protective Intelligence layer translates the project's country, city, access, and exposure indicators into a movement-level decision-support score. This layer is designed to reflect an intelligence-led executive protection approach: identify exposure before movement, evaluate local threat context, assess predictability and venue risk, and convert those inputs into a protective posture recommendation.

The highest Protective Intelligence priority in the current trip input set is TRIP001 for Executive A in Salvador, Brazil. The PI Risk Score is 87.09, with a signal of Severe PI Priority and posture recommendation: Senior Security Review / Consider Postponement.

Top Protective Intelligence Priorities

Rank	Trip	Principal	City	Country	Scenario	PI Score	PI Signal	Posture	Context
1	TRIP001	Executive A	Salvador	Brazil	Public event	87.09	Severe PI Priority	Senior Security Review / Consider Postponement	Matched city/location context
2	TRIP003	Executive C	Kherson	Ukraine	Site visit	78.1	High PI Priority	Route / Venue Redesign Recommended	Matched city/location context
3	TRIP002	Executive B	Tijuana	Mexico	Investor meeting	52.36	Moderate PI Priority	Enhanced Advance Work	Matched city/location context

Protective Intelligence Component Scores

Trip	Local Threat	Principal	Predictability	Venue/Airport	Online Leak	Support Gap	Rep./Sector	Base	Multiplier	PI Score
TRIP001	60.56	75.0	75.0	65.0	75.0	52.88	61.25	66.99	1.3	87.09
TRIP003	56.49	75.0	75.0	75.0	50.0	47.1	75.0	65.08	1.2	78.1
TRIP002	58.59	50.0	50.0	50.0	25.0	46.6	61.25	49.87	1.05	52.36

Analyst Priority Notes

Trip	Principal	City	Country	Priority Note
TRIP001	Executive A	Salvador	Brazil	Priority drivers: elevated local threat environment; high principal visibility; predictable movement pattern; online or itinerary exposure concern.
TRIP003	Executive C	Kherson	Ukraine	Priority drivers: high principal visibility; predictable movement pattern; elevated venue/airport/hotel exposure; heightened reputational or sector sensitivity.
TRIP002	Executive B	Tijuana	Mexico	Routine monitoring posture; no dominant exposure driver identified by the model.

Interpretation note: the Protective Intelligence score is a planning and triage indicator. It combines local threat environment, principal visibility, movement predictability, venue/airport/hotel exposure, online or itinerary exposure, support-access constraints, and reputational or business sensitivity. It does not replace protective advances, itinerary-specific threat assessment, law-enforcement liaison, venue security review, or real-time protective intelligence.

Protective Intelligence COA Decision-Support Layer

The Protective Intelligence COA decision-support layer converts trip-level risk scores and trigger flags into explainable courses of action for analyst review. The goal is to move beyond identifying elevated risk and toward documenting the type of review, validation, monitoring, or escalation that may be appropriate in a protective intelligence workflow.

The highest COA decision-support priority is TRIP001 for Executive A in Salvador, Brazil. The PI Risk Score is 87.09, with COA level: Senior Review and recommendation: Senior security review.

Top COA Decision-Support Priorities

Trip	Principal	City	Country	Scenario	PI Score	Band	COA	Recommendation	Primary Driver	Confidence
TRIP001	Executive A	Salvador	Brazil	Public event	87.09	Severe	Senior Review	Senior security review	severe aggregate protective intelligence score	High
TRIP003	Executive C	Kherson	Ukraine	Site visit	78.1	High	Security Lead Review	Notify security lead	high aggregate protective intelligence score	High
TRIP002	Executive B	Tijuana	Mexico	Investor meeting	52.36	Moderate-Elevated	Validate	Request analyst validation	elevated aggregate protective intelligence score	High

Decision Drivers and Analyst Notes

Trip	COA	Primary Driver	Secondary Driver	Supporting Indicators	Analyst Note
TRIP001	Senior Review	severe aggregate protective intelligence score	predictable or highly visible movement pattern	severe aggregate protective intelligence score; predictable or highly visible movement pattern	Severe protective intelligence score. Senior review is recommended before relying on routine posture. Key drivers: severe aggregate protective intelligence score; predictable or highly visible movement pattern.
TRIP003	Security Lead Review	high aggregate protective intelligence score	predictable or highly visible movement pattern	high aggregate protective intelligence score; predictable or highly visible movement pattern	High-risk condition identified. Notify the security lead and validate current assumptions. Key drivers: high aggregate protective intelligence score; predictable or highly visible movement pattern.
TRIP002	Validate	elevated aggregate protective intelligence score		elevated aggregate protective intelligence score	Analyst validation is recommended to confirm data recency, local context, and planning assumptions. Key drivers: elevated aggregate protective intelligence score.

Decision Trigger Flags

Trip	Triggers	Support Gap	Predictable	Online	Momentum	City Risk	High+Gap	Multi
TRIP001	2	False	True	False	False	False	False	False
TRIP003	1	False	True	False	False	False	False	False
TRIP002	0	False	False	False	False	False	False	False

Decision Rule Audit

Rule	Min	Max	Condition	COA	Recommendation	Rules Loaded	Records
COA-001	0	29	none	Routine	Continue routine monitoring	True	3
COA-002	30	44	moderate_score	Monitor	Increase monitoring	True	3
COA-003	45	59	elevated_score	Validate	Request analyst validation	True	3
COA-004	60	69	high_score	Enhanced Advance	Require enhanced advance work	True	3
COA-005	70	79	high_score_plus_support_gap	Security Lead Review	Notify security lead	True	3

Rule	Min	Max	Condition	COA	Recommendation	Rules Loaded	Records
COA-006	80	100	severe_score	Senior Review	Senior security review	True	3
COA-007	65	100	predictable_movement	Movement Review	Review route venue timing	True	3
COA-008	60	100	online_visibility	Information Exposure Review	Review online visibility and itinerary exposure	True	3
COA-009	70	100	city_risk_momentum	Protective Intelligence Watch	Place on protective intelligence watchlist	True	3
COA-010	80	100	multiple_high_triggers	Posture Reassessment	Consider postponement relocation or alternate plan	True	3

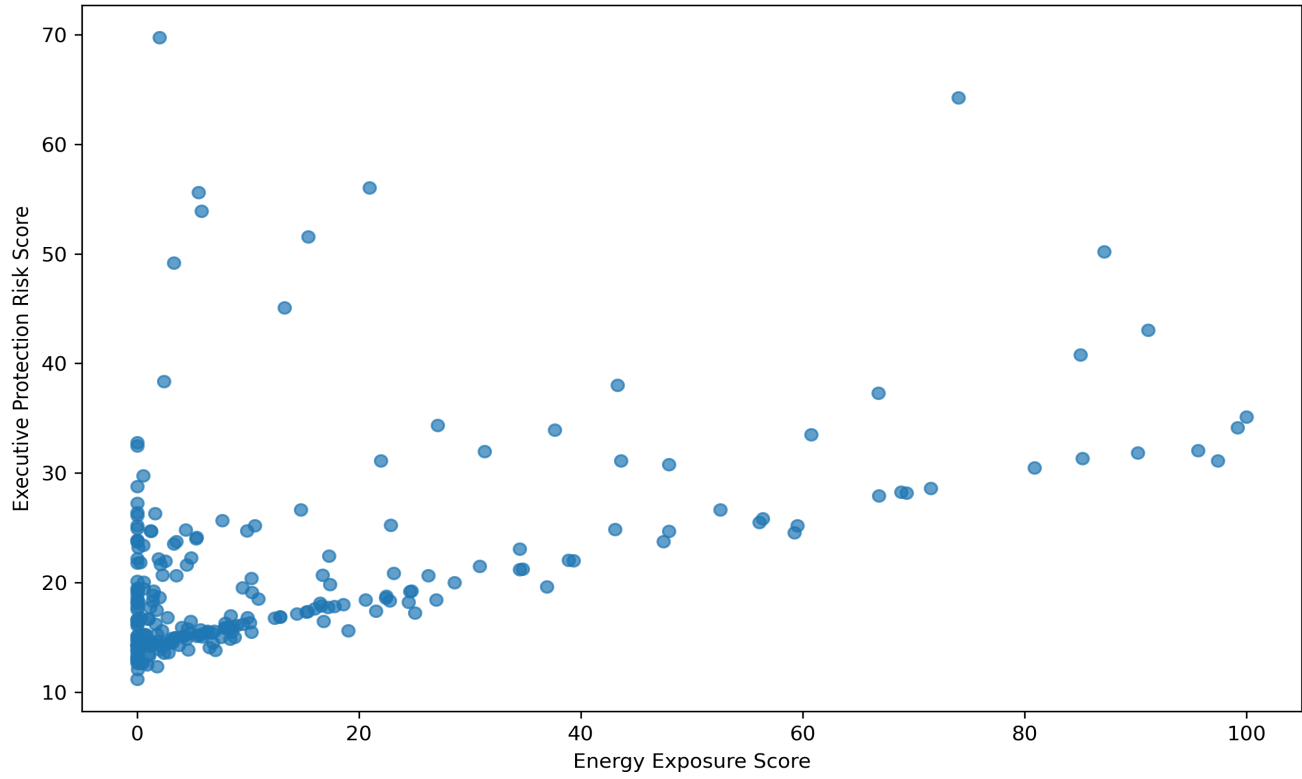
Interpretation note: the COA decision-support layer is an analyst-facing triage tool. It recommends review posture categories such as monitoring, validation, enhanced advance work, security lead review, senior review, or posture reassessment. It does not provide tactical instructions, route approval, venue approval, or a real-world incident probability forecast.

Energy Exposure, Governance, and Crime Risk

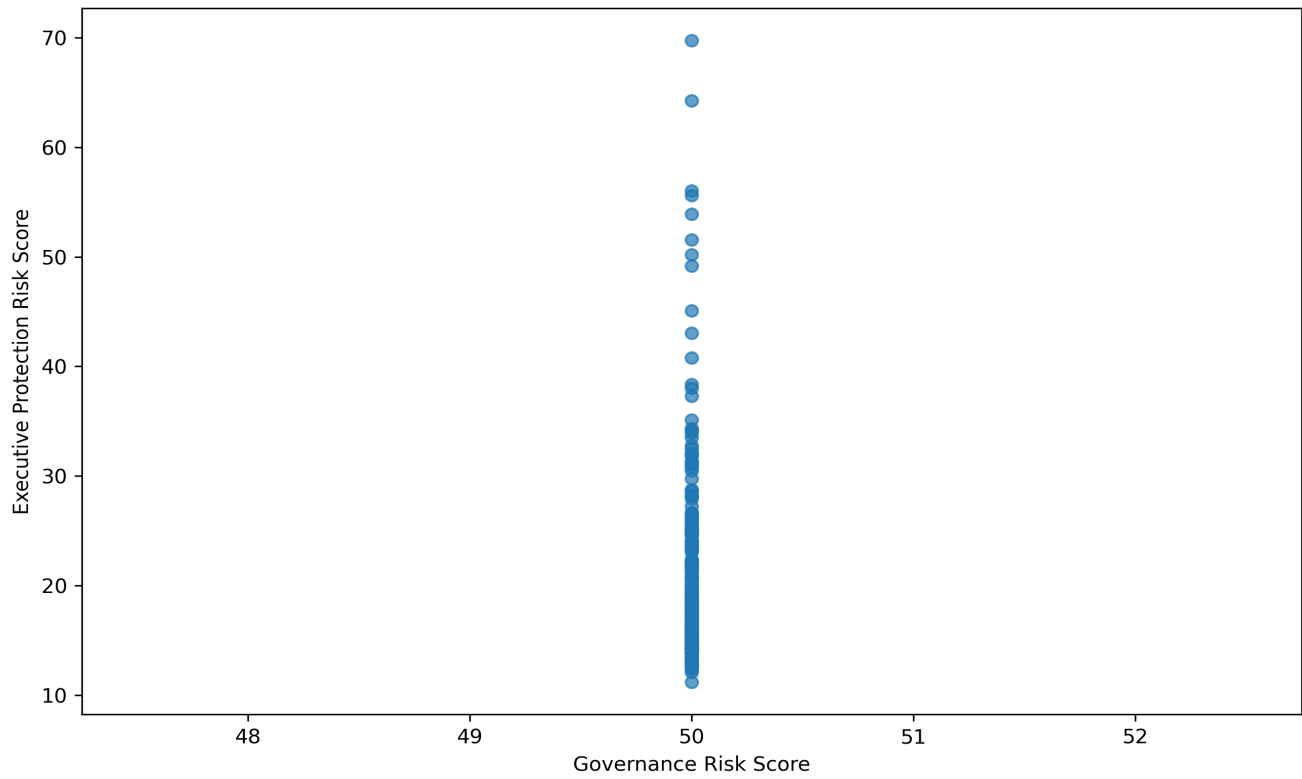
Energy exposure is included because executive protection risk for major energy firms is shaped not only by general country risk, but also by the strategic visibility of energy assets, infrastructure, exports, and market dependence. Governance risk captures institutional weakness, rule-of-law risk, corruption-control weakness, and political stability. The violent-crime proxy uses homicide rate per 100,000 people where available, including latest-available carry-forward logic when current-year homicide values lag.

Country	Homicide	Crime Yr	Crime Quality	Energy Raw	Energy Quality	Coverage
Ukraine	3.78	2021.0	Homicide proxy populated	1.18	Energy exposure populated	Strong data coverage
Nigeria	15.75	2023.0	Homicide proxy populated	39.89	Energy exposure populated	Strong data coverage
Myanmar	2.58	2023.0	Homicide proxy populated	11.46	Energy exposure populated	Strong data coverage
Mexico	24.86	2023.0	Homicide proxy populated	3.09	Energy exposure populated	Strong data coverage
Lebanon	2.24	2020.0	Homicide proxy populated	3.08	Energy exposure populated	Strong data coverage
Brazil	19.28	2023.0	Homicide proxy populated	8.39	Energy exposure populated	Strong data coverage
Oman	0.14	2023.0	Homicide proxy populated	48.41	Energy exposure populated	Strong data coverage
Sudan	2.58		Median-filled homicide proxy	1.98	Energy exposure populated	Moderate data coverage
India	2.82	2022.0	Homicide proxy populated	7.08	Energy exposure populated	Strong data coverage
Guyana	19.12	2023.0	Homicide proxy populated	50.01	Energy exposure populated	Strong data coverage

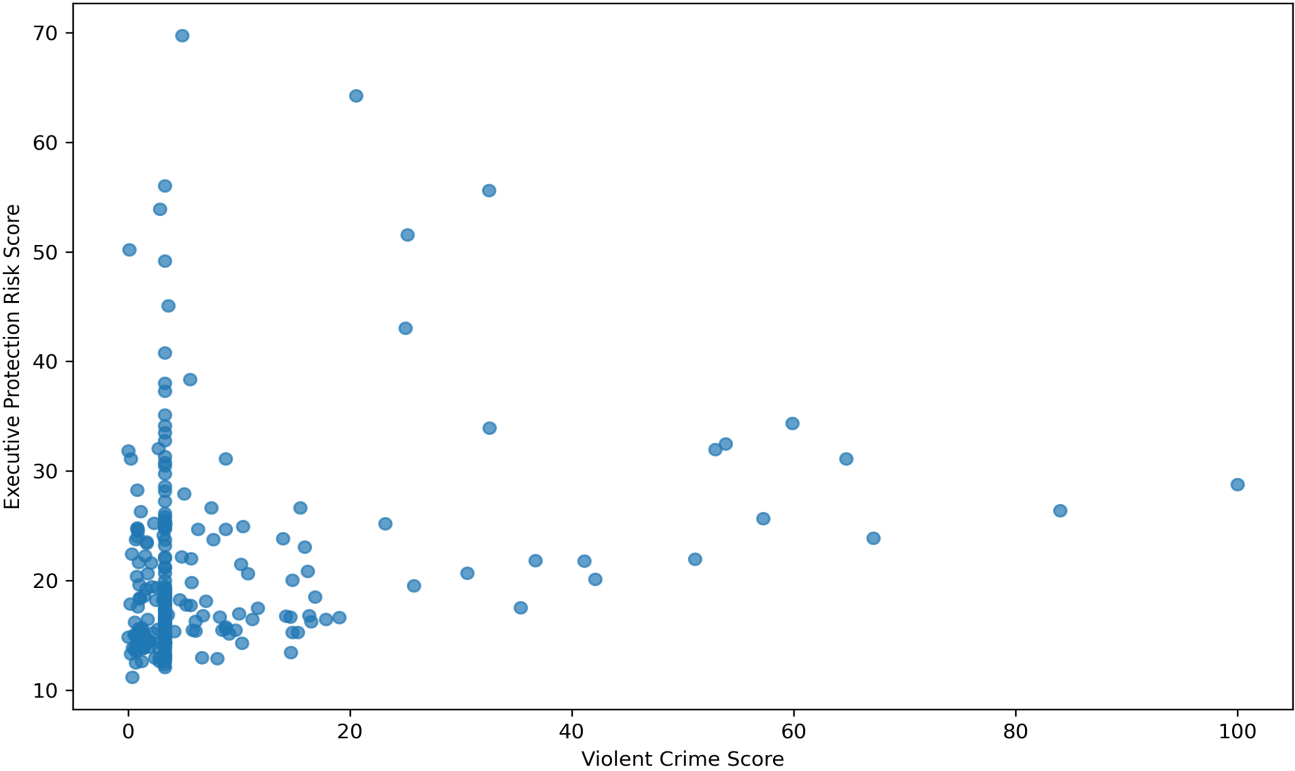
Energy Exposure vs. Executive Protection Risk



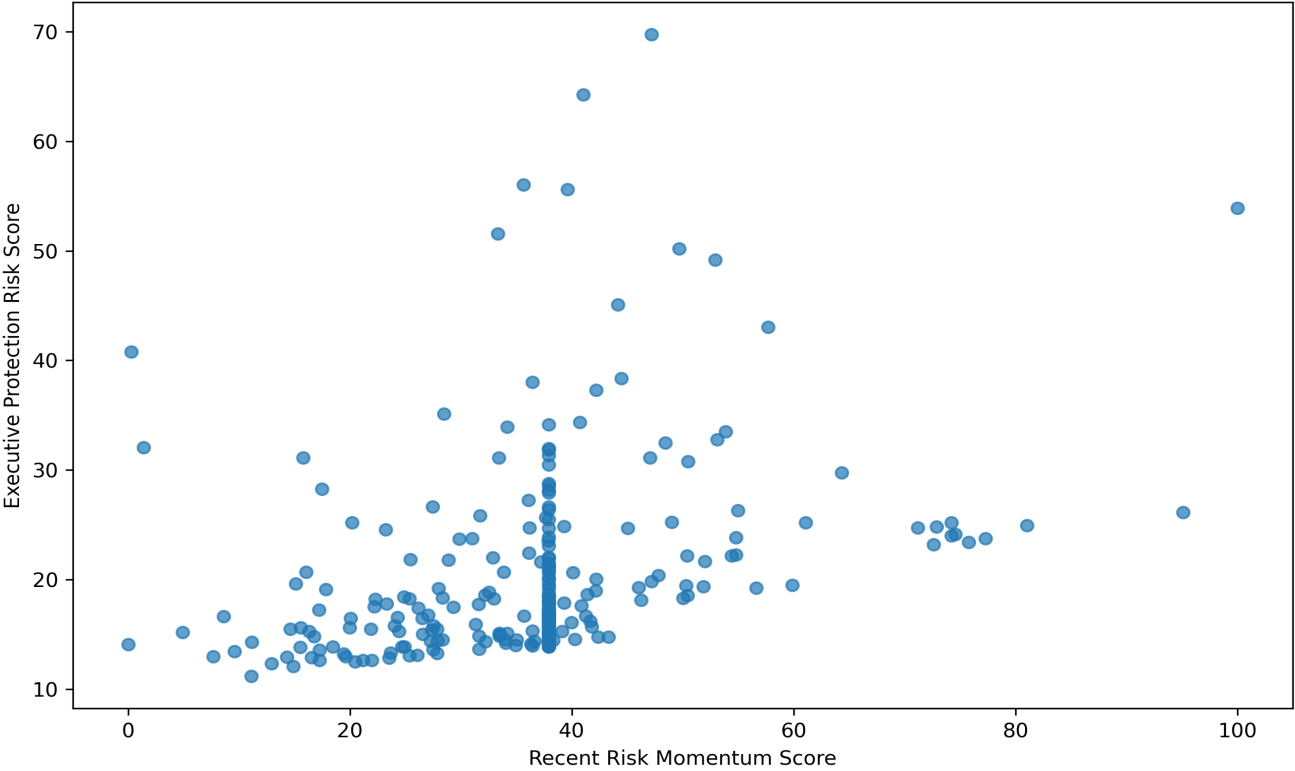
Governance Risk vs. Executive Protection Risk



Violent Crime Proxy vs. Executive Protection Risk



Recent Risk Momentum vs. Executive Protection Risk



Scenario Analysis

Scenario multipliers estimate how baseline country risk may change under specific executive protection contexts, such as public energy events, site visits to energy assets, travel during civil unrest, major project announcements, or high-visibility executive visits.

Scenario Summary

Scenario	Avg.	Max	Elev+	High	Severe
high_visibility_executive_visit	28.57	97.66	14	5	2
travel_during_civil_unrest	27.55	94.18	14	3	2
major_energy_project_announcement	26.53	90.69	11	4	1
labor_unrest_or_protest_environment	25.51	87.2	11	2	1
site_visit_to_energy_asset	24.49	83.71	10	2	0
public_energy_event	23.47	80.22	9	2	0
routine_executive_travel	20.41	69.76	7	0	0

High-Visibility Executive Visit: Top Countries

Rank	Country	Base	Scenario	Lift	Bucket
1	Ukraine	69.76	97.66	27.9	Severe
2	Nigeria	64.25	89.95	25.7	Severe
3	Myanmar	56.04	78.46	22.42	High
4	Mexico	55.61	77.85	22.24	High
5	Lebanon	53.9	75.46	21.56	High
6	Brazil	51.55	72.17	20.62	High
7	Oman	50.18	70.25	20.07	High
8	Sudan	49.18	68.85	19.67	Elevated
9	India	45.08	63.11	18.03	Elevated
10	Guyana	43.03	60.24	17.21	Elevated

Sensitivity Analysis

Sensitivity analysis recalculates country rankings under alternative weighting assumptions. This helps assess whether the top-risk countries remain elevated because of broad-based risk exposure or because of one dominant model component.

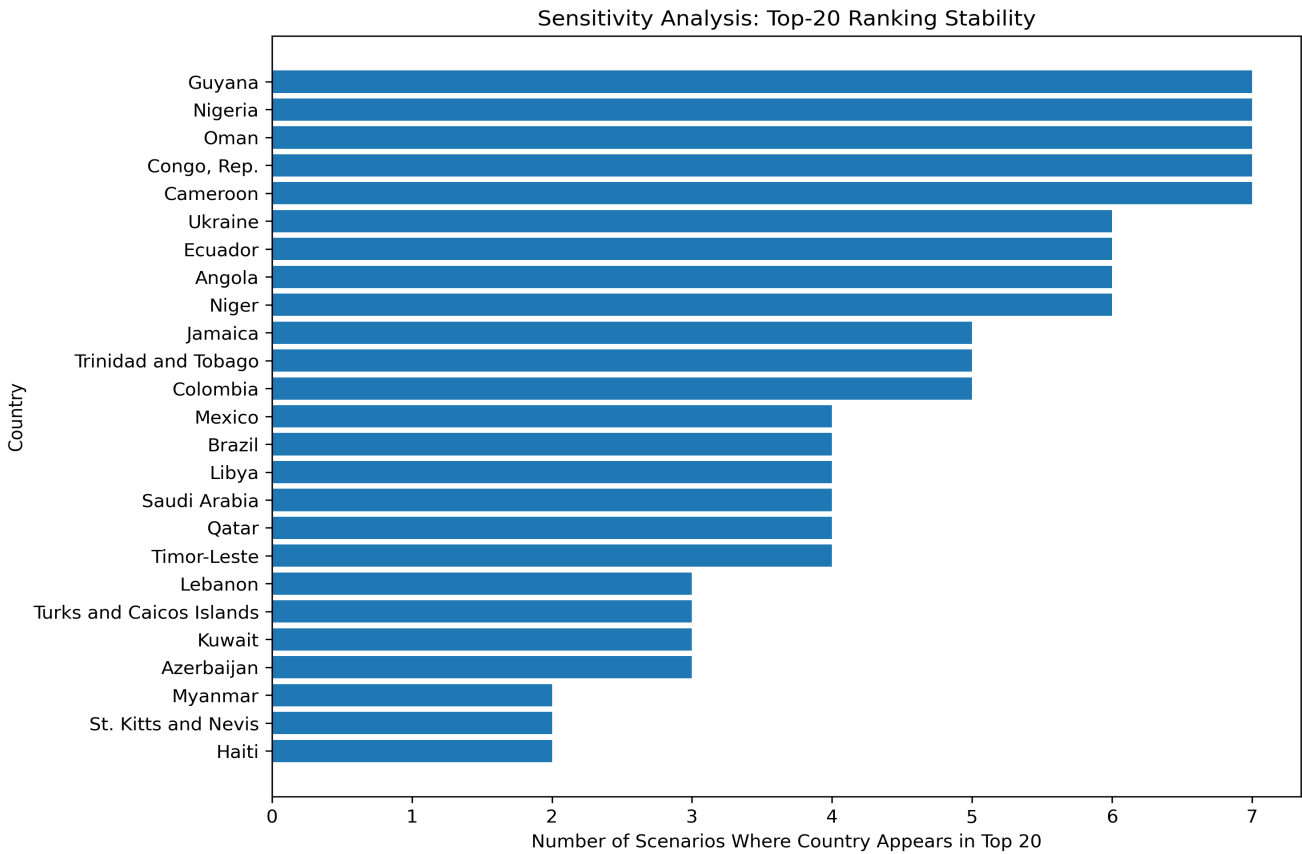
Scenarios	Any Top 20	All Scenarios	Majority	Highly Stable	Stable+	Most Sensitive	Avg Rank Chg.
7	48	5	18	1	9	recent_momentum_heavy	28.9

The top-risk group is meaningfully sensitive to weighting assumptions; interpret exact ranks with caution.

Most Stable Top-20 Countries

Country	Top20 Count	Share	Best	Worst	Avg Rank	Stability
Guyana	7	1.0	1	9	3.71	Stable
Nigeria	7	1.0	2	7	3.71	Highly Stable
Oman	7	1.0	3	16	5.86	Stable
Congo, Rep.	7	1.0	3	18	9.86	Stable

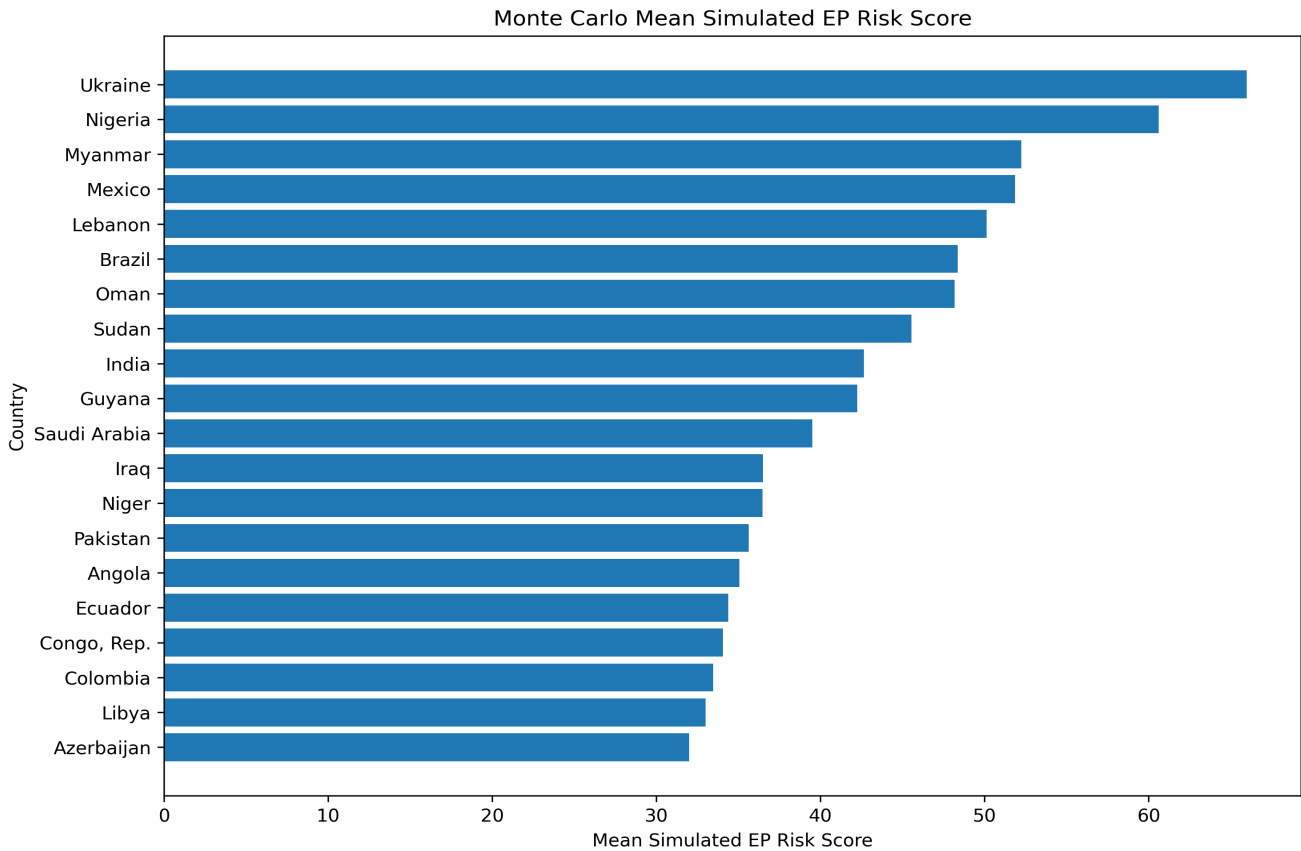
Country	Top20 Count	Share	Best	Worst	Avg Rank	Stability
Cameroon	7	1.0	11	20	16.14	Stable
Ukraine	6	0.86	1	8	3.5	Stable
Ecuador	6	0.86	2	9	5.5	Stable
Angola	6	0.86	2	17	8.17	Stable
Niger	6	0.86	9	15	12.67	Stable
Jamaica	5	0.71	4	19	11.2	Moderately Sensitive
Trinidad and Tobago	5	0.71	5	16	9.6	Moderately Sensitive
Colombia	5	0.71	12	18	15.2	Moderately Sensitive

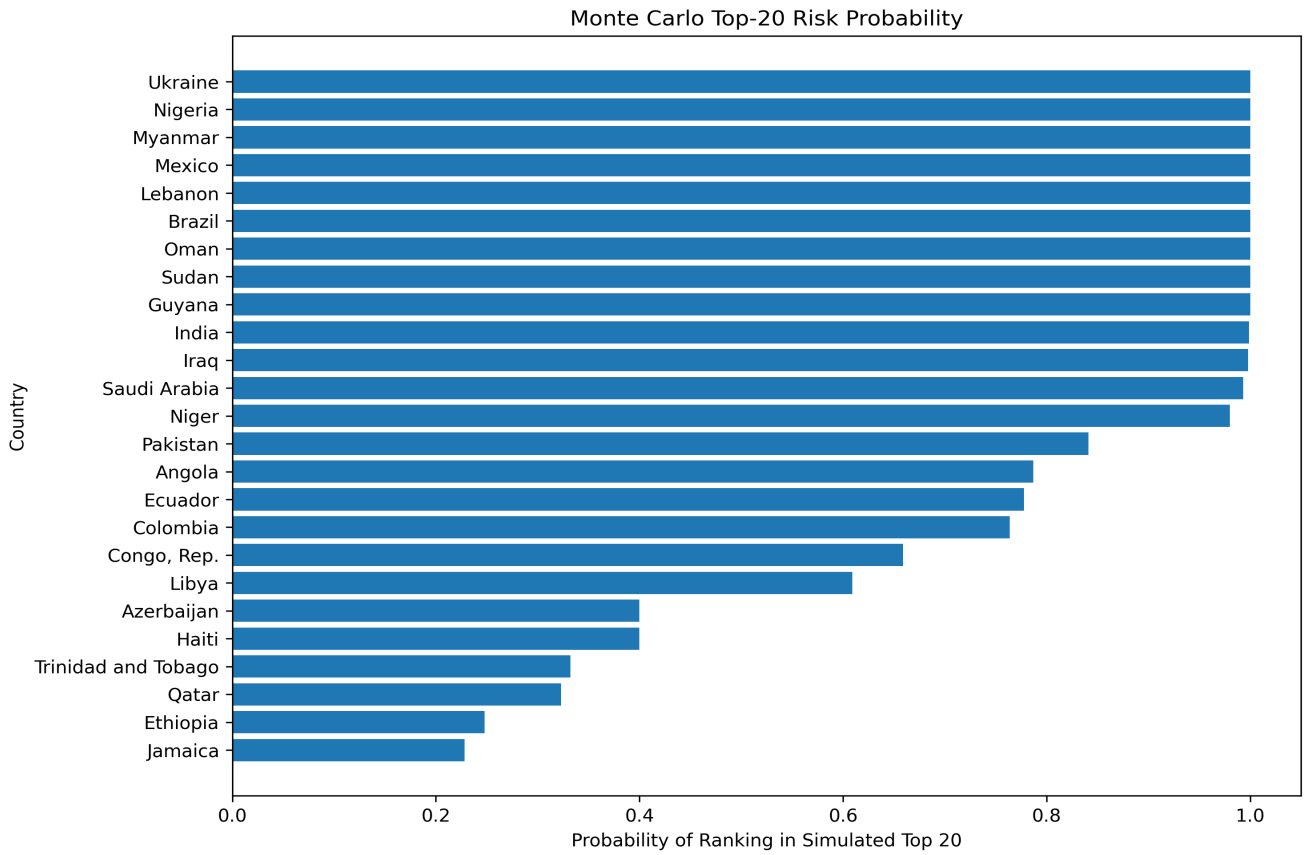


Monte Carlo Risk Simulation

Monte Carlo simulation tests how stable country risk rankings remain when model component weights are randomly perturbed around the baseline weighting structure. This helps identify countries that remain high-risk across many plausible weighting assumptions rather than only under one deterministic model setup.

The simulation preserves the model's core components and bounded severity calibration concept, while allowing civil unrest, governance, violent crime, energy exposure, and recent momentum weights to vary. Countries with high top-20 probability are more robustly classified as priority-risk countries.





Most Monte Carlo-Stable Top-Risk Countries

Country	Base Rank	Base	Bucket	Mean Sim.	Vol.	Mean Rank	Top20 Prob.	Stability
Ukraine	1	69.76	Elevated	65.99	3.8	1.16	1.0	Highly Stable Top-20 Risk
Nigeria	2	64.25	Elevated	60.61	2.22	1.85	1.0	Highly Stable Top-20 Risk
Myanmar	3	56.04	Elevated	52.25	1.74	3.73	1.0	Highly Stable Top-20 Risk
Mexico	4	55.61	Elevated	51.86	1.62	4.12	1.0	Highly Stable Top-20 Risk
Lebanon	5	53.9	Elevated	50.13	3.25	5.19	1.0	Highly Stable Top-20 Risk
Brazil	6	51.55	Elevated	48.38	1.28	6.31	1.0	Highly Stable Top-20 Risk
Oman	7	50.18	Elevated	48.17	3.42	6.24	1.0	Highly Stable Top-20 Risk
Sudan	8	49.18	Moderate	45.56	2.19	8.13	1.0	Highly Stable Top-20 Risk
Guyana	10	43.03	Moderate	42.24	3.76	9.37	1.0	Highly Stable Top-20 Risk
India	9	45.08	Moderate	42.65	1.88	9.8	1.0	Highly Stable Top-20 Risk
Iraq	13	38.03	Moderate	36.49	1.94	13.8	1.0	Highly Stable Top-20 Risk
Saudi Arabia	11	40.78	Moderate	39.51	3.42	11.14	0.99	Highly Stable Top-20 Risk

Countries Most Sensitive to Weight Uncertainty

Country	Base Rank	Base	Mean Sim.	Vol.	Score Range	Rank Range	Top20 Prob.
Kuwait	28	31.1	31.03	4.22	26.46	79	0.22

Country	Base Rank	Base	Mean Sim.	Vol.	Score Range	Rank Range	Top20 Prob.
Congo, Rep.	17	34.13	34.06	4.21	26.91	46	0.66
Angola	15	35.12	35.05	4.08	25.71	40	0.79
Turks and Caicos Islands	32	28.79	28.85	4.03	25.86	105	0.15
Azerbaijan	22	32.06	31.99	4.02	25.15	69	0.4
Qatar	24	31.83	31.77	3.92	25.43	50	0.32
Ukraine	1	69.76	65.99	3.8	22.61	2	1.0
Guyana	10	43.03	42.24	3.76	22.89	12	1.0

Monte Carlo results should be interpreted as a robustness test of model assumptions, not as a probability forecast of real-world security events.

Model Governance and Methodology Controls

This section documents the model's intended use, data-source dependencies, scoring assumptions, scenario overlays, sensitivity design, forward-risk guardrails, and limitations. It is intended to make the framework more transparent, auditable, and suitable for portfolio presentation.

Governance Summary

Area	Description	Model Implication
Project purpose	Independent OSINT research model for country-level executive protection risk screening in global energy operations.	Designed for strategic screening, portfolio demonstration, and risk-intelligence research.
Intended use	Country-level prioritization for executive travel, site visits, public events, and energy-sector operating-environment review.	Useful for early-stage triage and comparative country risk analysis.
Not intended use	Not a tactical travel approval tool, protective advance plan, route-security assessment, or replacement for local intelligence.	Scores should not be used as standalone operational decisions.
Data sources	Uses public ACLED event data, World Bank governance and macro indicators, World Bank homicide proxy data, and World Bank energy exposure indicators.	Results depend on public-data availability, reporting coverage, indicator lag, and country-name matching.
Scoring structure	Composite score uses civil unrest / political violence, governance risk, violent crime proxy risk, energy exposure, and recent risk momentum.	Countries can rank highly because of conflict severity, weak governance, crime exposure, energy relevance, or momentum.
Severity calibration	A bounded uplift is applied for extreme conflict exposure, fatality severity, geographic spread, compound governance/violence risk, energy instability, and momentum.	Prevents severe environments from being compressed into moderate scores by pure weighted averaging.
Scenario analysis	Scenario multipliers estimate how baseline risk may change under operational contexts such as public events, site visits, civil unrest, or high-visibility executive travel.	Scenario outputs are planning overlays, not deterministic forecasts.
Sensitivity analysis	Alternative weighting scenarios test how stable country rankings are under different model assumptions.	Countries stable across scenarios are more robust top-risk candidates.
Forward-risk layer	Forward update compares 2026 ACLED activity against 2025 and anchors results to the 2024 baseline.	If target-year ACLED data is unavailable, the model retains the baseline rather than treating missing data as improvement.
Affiliation disclaimer	Independent portfolio research project. Not affiliated with, endorsed by, or representative of any company, government agency, security organization, ACLED, the World Bank, or other referenced organization.	The model should be interpreted as an open-source analytical framework.

Key Model Assumptions

Type	Assumption	Value	Notes
Baseline risk weight	civil_unrest_political_violence_score	0.35	Component weight used in the baseline EP risk score.
Baseline risk weight	governance_risk_score	0.2	Component weight used in the baseline EP risk score.
Baseline risk weight	violent_crime_score	0.15	Component weight used in the baseline EP risk score.
Baseline risk weight	energy_exposure_score	0.2	Component weight used in the baseline EP risk score.
Baseline risk weight	recent_risk_momentum_score	0.1	Component weight used in the baseline EP risk score.
Risk bucket	Low	0 to 25	Risk bucket threshold used for score interpretation.
Risk bucket	Moderate	25 to 50	Risk bucket threshold used for score interpretation.
Risk bucket	Elevated	50 to 70	Risk bucket threshold used for score interpretation.
Risk bucket	High	70 to 85	Risk bucket threshold used for score interpretation.
Risk bucket	Severe	85 to 101	Risk bucket threshold used for score interpretation.
Scenario multiplier	routine_executive_travel	1.0	Multiplier applied to baseline EP risk score.
Scenario multiplier	public_energy_event	1.15	Multiplier applied to baseline EP risk score.
Scenario multiplier	site_visit_to_energy_asset	1.2	Multiplier applied to baseline EP risk score.
Scenario multiplier	travel_during_civil_unrest	1.35	Multiplier applied to baseline EP risk score.

Governance note: the model is designed as an open-source strategic screening framework. It does not replace local protective intelligence, itinerary-specific threat assessment, route analysis, venue security review, or real-time monitoring.

Model Diagnostics and Data Coverage

Diagnostics summarize file availability, component maturity, missing-value coverage, baseline model maturity, and forward-layer availability.

Model Maturity

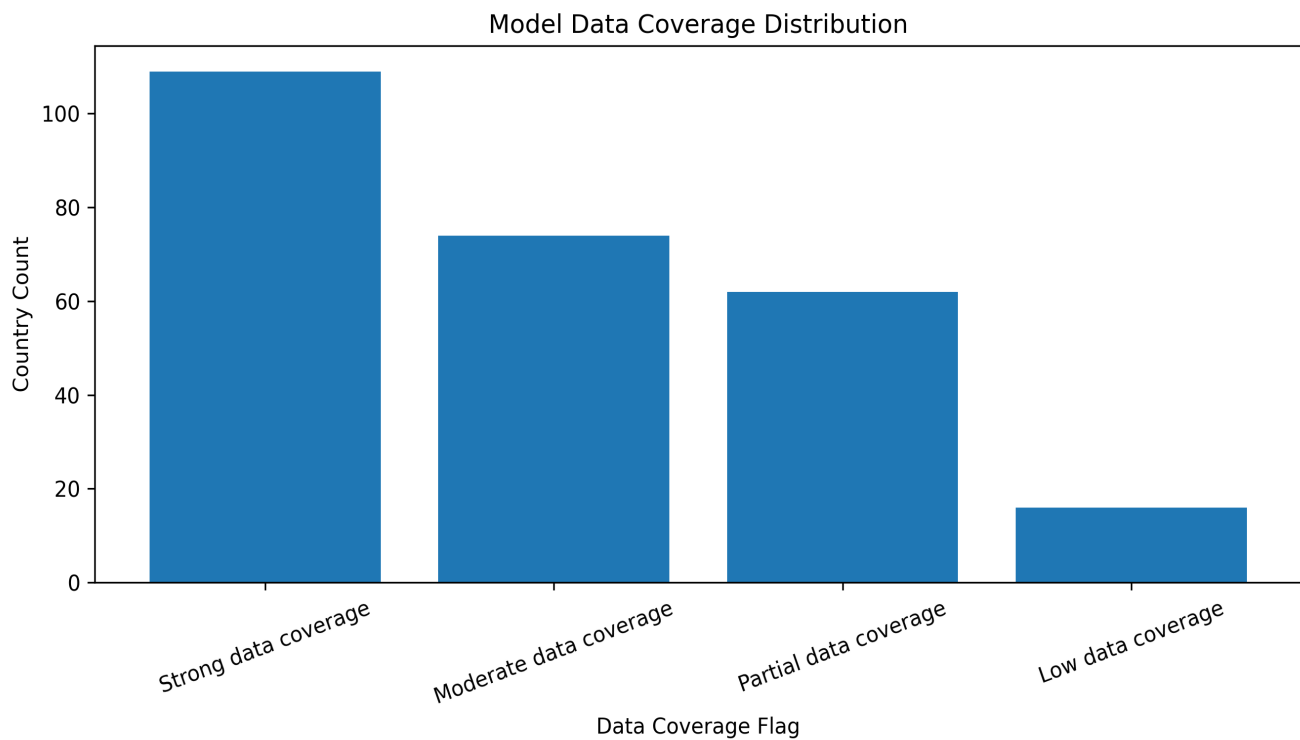
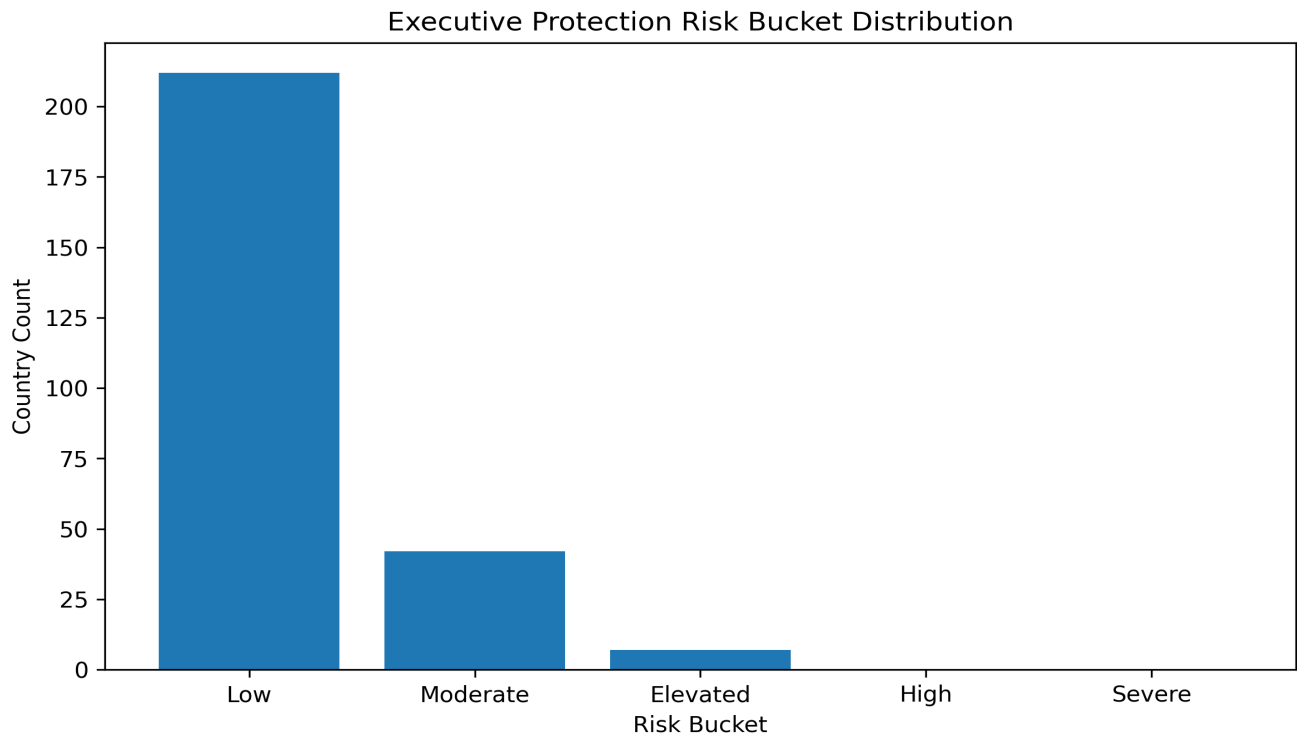
Maturity	Live	Total	Placeholder	Placeholder Share	Forward Layer
Fully Data-Driven	5	5	0	0.0	Forward layer initialized / baseline retained; target-year ACLED data unavailable

Component Maturity

Component	Group	Status	Rows	Placeholder	Flag
Governance / macro risk	Baseline model	Live Data	261	False	Component populated.
Civil unrest / political violence risk	Baseline model	Live Data	1080	False	Component populated.
Violent crime proxy risk	Baseline model	Live Data	755	False	Component populated.
Energy-sector exposure risk	Baseline model	Live Data	1305	False	Component populated.
Final model output	Baseline model	Live Data	261	False	Component populated.
2026 forward ACLED trend layer	Forward nowcast layer	Live Data	20	False	Forward layer initialized; target-year ACLED unavailable and forward scores are baseline-retained.
2026 forward risk output	Forward nowcast layer	Live Data	20	False	Forward layer initialized; target-year ACLED unavailable and forward scores are baseline-retained.

Component Coverage Summary

Area	Countries	Metric	Value	Flag
World Bank indicators	261	country_code count	261	World Bank indicator file populated
ACLED raw events	234	raw event rows	1509322	ACLED raw event file populated
ACLED country-year features	234	countries with country-year ACLED features	234	ACLED processed features populated
ACLED event coverage	234	total processed ACLED events	1509322	ACLED event counts available
ACLED geospatial coverage	234	country-years with coordinate spread populated	1080	Coordinate-pair spread feature available
Homicide proxy	151	rows with populated homicide proxy	151	Crime proxy populated
Homicide proxy quality flags	151	Homicide proxy missing	604	Crime data quality distribution
Homicide proxy quality flags	151	Homicide proxy populated	151	Crime data quality distribution
Energy exposure	261	rows with measurable energy exposure	215	Energy exposure features populated
Energy exposure quality flags	261	No measurable energy exposure	1090	Energy data quality distribution
Energy exposure quality flags	261	Energy exposure populated	215	Energy data quality distribution
Master dataset coverage flags	261	Strong data coverage	109	Master data coverage flag distribution



Highest Missing-Value Columns

column	missing_count	missing_share	non_missing_count
homicide_rate_per_100k_year	110	0.42	151
homicide_rate_per_100k_indicator_code	110	0.42	151

column	missing_count	missing_share	non_missing_count
crime_data_source	110	0.42	151
acled_data_source	90	0.34	171
fuel_exports_pct_merchandise_exports_year	53	0.2	208
fuel_exports_pct_merchandise_exports_indicator_code	53	0.2	208
oil_rents_pct_gdp_year	22	0.08	239
oil_rents_pct_gdp_indicator_code	22	0.08	239
natural_gas_rents_pct_gdp_year	22	0.08	239
natural_gas_rents_pct_gdp_indicator_code	22	0.08	239

Limitations

This project is an independent research and portfolio project. It is not affiliated with, endorsed by, or representative of any company, government agency, security organization, ACLED, the World Bank, or any other organization referenced through public data sources.

This model uses public country-level and event-level data. It does not include confidential travel itineraries, executive profiles, company asset-level security data, local protective intelligence, law-enforcement liaison information, proprietary threat reporting, or route-specific protective advances. Therefore, it should be viewed as a strategic screening model rather than a tactical protection plan.

Country-level scores can obscure local variation. A low-risk national score does not mean that every city, venue, route, event, or asset is low risk. Conversely, a high-risk national score does not mean that all travel is infeasible. The model is best used as a prioritization and analytical triage tool.

The 2026 Forward Risk Update is a nowcast-style analytical layer designed to use recent ACLED activity for the current top-risk countries. In runs where 2026 ACLED data is unavailable, the model retains the calibrated baseline rather than treating missing data as risk improvement. It should not be interpreted as a tactical forecast or as a replacement for local protective intelligence, itinerary-specific threat assessment, or real-time security monitoring.

The airport and medical access layer uses public proxy data. Airport distance is based on reference airport locations, and medical capacity is based on country-level World Bank indicators. These features should be interpreted as planning-support context, not as a medical, evacuation, logistics, or tactical security assessment.

The Protective Intelligence posture layer uses user-supplied trip and exposure assumptions. Its recommendations should be interpreted as a decision-support prioritization aid, not as a final travel approval, route plan, venue advance, family exposure review, or no-go determination.

The COA decision-support layer translates scores and trigger flags into review recommendations for analyst triage. These outputs should not be treated as binding operational decisions, tactical instructions, route approval, venue approval, protective detail staffing guidance, or final go/no-go determinations.